ESSnet Big Data

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Work Package 7

Multi domains

Deliverable:

7.1 Report for Population domain

7.2 Report for Tourism/Border crossing

7.3 Report for Agriculture

This report contains basic information on data access (such as legal and privacy aspects), data quality issues, methodology (including the combination of data) and the technical aspects of the data.

Prepared by WP 7 team

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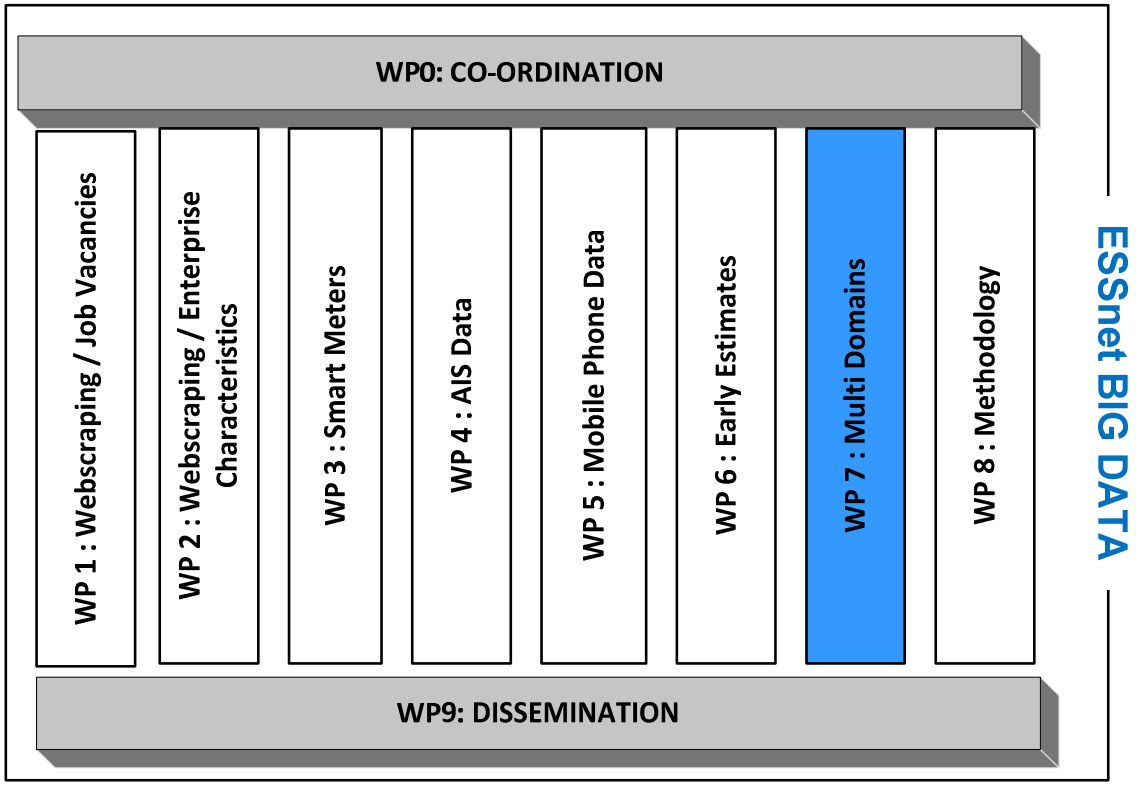
Name/abbreviation	*Explanation*
BDAR	Big Data Action Plan and Roadmap
FPA	Framework Partnership Agreement
SGA	Specific Grant Agreement
ESSnet	A network of several European Statistical System (ESS) aimed at providing results that will be beneficial to the whole ESS.
ESSC	European Statistical System Committee. The task of the Committee is to "provide professional guidance to the ESS for developing, producing and disseminating European statistics" (article 7 of the Regulation).
ESS	European Statistical System. This is the partnership between the Community statistical authority, which is the Commission (Eurostat), and the national statistical institutes (NSIs) and other national authorities responsible in each Member State for the development, production and dissemination of European statistics.
WP	Work package
GDDKiA	the General Directorate of National Roads and Motorways in Poland
GUS	Statistics Poland
CBS	Statistics Netherlands
CSO	Statistics Ireland
ONS	Statistics UK
BD	Big Data

Work package 7. Multi domains - introduction

Work package number seven is one of the ESSnet Big Data teams.

The ESSnet Big Data Project is a part of the Big Data Action Plan and Roadmap 1.06, which was agreed to be to integrated into the ESS Vision 2020 portfolio. The overall objective of the project is to prepare the ESS for integration of Big Data sources into the production of official statistics. The award criteria mentioned that the project has to focus on running pilot projects exploring the potential of selected Big Data sources for producing or contributing to the production of official statistics. The aim of these pilots is to undertake concrete action in the domain of Big Data and obtain hands-on experience in the use of Big Data for official statistics.

A consortium of 22 partners, consisting of 20 National Statistical Institutes and 2 Statistical Authorities, has been formed in September 2016 to meet the objectives of the project. According to the Framework Partnership Agreement (FPA) between the consortium and Eurostat, the project runs between February 2016 and May 2018. To concentrate the work as much as possible on the pilots, the consortium has organized its work around the pilots. More specifically, the consortium has subdivided its work into work packages (WP).



The aim of WP 7 is to find out how a combination of Big Data sources, administrative data and statistical data may enrich statistical output. In other words, investigate how they can be used to improve current statistics in the domains: 'Population', 'Tourism/border crossings' and 'Agriculture'.

This work package is scientific in nature. From a methodological, qualitative and technical point of view, it is required to work with professional independence. The activities in this work package are independent from the results of the work packages 1 to 5, although relevant information with other work packages will be exchanged. One such example of this is WP 6 and WP 7 whose aims are related. These WPs both concern the exploration of combining various sources, including Big Data sources, for the production of statistics.

The WP team will describe the data collection, linkage, processing and methodology, when combining this data in statistical domains. Additional value could be discovered in the linkages between domains.

Since this WP considers many cross-cutting issues, such as methodology, quality and technical requirements, its outputs can be used as inputs for the WP 8 of SGA-2. The participation of NL, which is expected to lead WP 8 of SGA-2, is aimed at ensuring this data is used to its full capability. Apart from GUS (Statistics Poland) which is leading WP 7 and CBS (Statistics Netherlands), this WP is carried out by two other representatives of: CSO (Statistics Ireland) and ONS (Statistics United Kingdom). From SGA-2 (in March 2017) Portugal will join to this team.

WP 7 team divided work into 4 main groups of tasks:

- Task 1. Data availability/Data inventory (SGA-1)
- Task 2. Data feasibility (SGA-1)
- Task 3. Data combination (SGA-2)
- Task 4. Summary plus future perspectives (SGA-2)

Similarities and differences between countries, concerning the availability of registers, and the legality of data linkage will be taken into account when carrying out the tasks.

The first step and the basis for further works was to prepare data inventory. This is a wide list of data sources. It was created regardless of whether data are available or not. Conducting brainstorming sessions was crucial to implement this activity. In Poland WP 7 team have conducted 6 sessions – one in a face-to-face form took place during a meeting on 26/04/2016, the remaining 5 were conducted in electronic form. The collected information has been mapped to the UNECE typology of Big Data sources, which enable researchers to organize the data, group the data, determine the level of detailed description and of suitability of the data. Foreign partners (UK, Ireland and the Netherlands) have received a request for a similar brainstorming session and preparation of preliminary lists of sources. The results of all sessions have been described in first WP

7 Milestone: Progress and technical report of internal WP-meeting (document is available at the link https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/index.php/WP 7_2016_05 _24_Webex).

At the beginning of June WP 7 team prepared (with WP 6 cooperation) and sent a questionnaire to participants (both domestic and foreign). The questionnaire was sent in the form of a url link online. The questions were focused on issues such as:

- potential of Big Data sources,
- using Big Data by statistical institutions,
- obstacles of using Big Data,
- access and legal aspects,
- IT,
- quality,
- methodology,
- combining different sources.

19 questionnaires were completed within the given time frame which provided significant information on the themes above. The detailed results were presented in the second Milestone 7.5 "List of available Big Data sources in the domain(s)" (Annex I).

These actions' (brainstorming and questionnaire) were intended to create the widest possible range of Big Data sources, so that in the later stages these sources could be verified from different points of view. Gradually the least useful sources were eliminated.

A valuable step was the WP 6 and WP 7 common meeting that took place in Warsaw in June 2016. This meeting's purpose was to exchange information and experience in using Big Data sources and arrangements for future work. An important point of the meeting was also to prepare and establish a framework for cooperation in SGA-2. This meeting involved a discussion which analysed the possibility of using sources for each domain (legal aspects, availability, methodology, IT and quality were investigated). As a result WP 7 team built the list of potential sources for each domain. It was at this meeting the initial ideas for use cases appeared. Now the work is focused mainly on the use cases. Their detailed description is presented in the next section~~s~~.

Domain	Population	Agriculture	Tourism/Border Crossing
Name of the use case	Everyday citizen satisfaction. Opinions about public events with major impact on peoples' satisfaction	Estimation of Agricultural statistics – pilot case study on crop types based on satellite data	Border movement

Big Data source	Social media/blogs/Internet portals	Satellite images	Traffic sensors
Responsibility	UK + PL, PT	PL+ IE	PL+ NL, PT (from SGA-2)
Brief overview of the methodology	Webscraping Data/Text/Web mining Machine learning	Combinations of data – data fusion on radar and optical remote sensing data; data comparison with traditional surveys e.g. FSS; combining data – administrative data sources with satellite data.	Intertemporal disaggregation and interpolation, Latent variable models, Cross entropy econometrics.

Deliverable 7.1. Report on *Population* domain area containing basic information on the data access, quality issues, methodology (focus on combining data) and technical aspects.

7.1.1 Background

Population is one of the few domains that have been subjected to statistical surveys. There are quantitative aspects like composition, density, distribution, growth, movement, size, and structure of the population. At the other end of the spectrum there are qualitative aspects (the sociological factors) such as education quality, crime, development, diet and nutrition, race, social class, wealth, wellbeing. A potential data source in these areas could be Big Data. Data enriched from various sources with additional details, as compared to what traditional systems offered, can deliver relevant and timely insights. The wide usage of social media in everyday life has made the world a close-knit circle.

Data published by the internet users allows the diagnosis of social moods[1]. These are the moods of people associated with various public events[2] or to observe the impact of public moods on a decision. It could be possible to analyse the moods of different countries' citizens in relation to the same events and examine mood changes over time.

Arguably, this data combines subjective opinion[3] and expectations concerning the quality of life in the country. They are formed by individual and environmental standards and aspirations. To some extent these standards correspond to objective indicators describing the socio-economic reality of life in the country. Although these are subjective sources, the emotion in the specific moment exists as an objective phenomenon and includes the potential to impact on the course of social and economic processes in real time[4].

There are difficulties in collecting on-line data. Data from social networks are generally unstructured and appear irregularly. Ethical aspects must also be taken into consideration. It is challenging to read statements written online and to identify the context of the phrase. Furthermore this data doesn't reflect the mood of all the population, but only of the people who are active in social networks.

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[1] Lansdall-Welfare T., Lampos V., Cristianini N., Nowcasting the mood of the nation, http://onlinelibrary.wiley.com/doi/10.1111/j.1740-9713.2012.00588.x/epdf

[2] Gruzd A., Doiron S., Mai P., Is Happiness Contagious Online? A Case of Twitter and the 2010 Olympics; https://www.researchgate.net/profile/Anatoliy_Gruzd/publication/215790760_Is_Happiness_Contagious_Online_A_Case_of_Twitter_and_the_2010_Winter_Olympics/links/543fd8090cf21227a11b8bb7.pdf

[3] O'Connor B., Balasubramanyan R., Routledge B.R., Smith N.A., From Tweets to Polls: Linking Text Sentiment to Public Opinion Time Series; http://homes.cs.washington.edu/~nasmith/papers/oconnor+balasubramanyan+routledge+smith.icwsm10.pdf

[4] Daas P.J.H., Puts M.J.H., Social Media Sentiment and Consumer Confidence, https://www.ecb.europa.eu/pub/pdf/scpsps/ecbsp5.en.pdf

It is possible the analysis of the people's moods made through Big Data can support and enrich traditional public opinion research. Arguably, the analysis of the people's moods derived from Big Data could potentially be more accurate than the previous traditional methods.

The main benefits and value added for official statistics: support traditional European Social Surveys and related to this, supplement the research methodology of some phenomena that are difficult to measure through traditional polls.

General tasks of WP 7 (population) are:

1. Data access (SGA-1) - Identifying and analyzing list of exploitable sources for population (legal aspects, availability, methodology, IT, quality)
2. Data feasibility (SGA-1) Selection the most value Big Data sources for population. Preliminary assessment of the usefulness - developing the assessment factors.
3. Data combination (SGA-2)
4. Summary plus future perspectives (SGA-2)

Several portals/ websites were taken into account, e.g. Facebook, Twitter, Instagram, Google Trends. Data quality was assessed by rating selected dimensions of the data quality, e.g.: complexity, completeness, usability, timeliness and frequency, accuracy, coherence, validity, accessibility, clarity, and so on, let us decide that Twitter is the best data source, and will be used for sure.

Population will be limited to:

- Residence population (and migration);
- Number of women per 100 men;
- Population structure.

It will be connected with social statistics for instance: satisfaction, moods and morbidity areas.

The following tasks are necessary:

- Choose the sources (e.g. Twitter, website portals) on which it will be possible to geolocate comments;
- Choose the topics (e.g. depression, Brexit) to explore people's opinion;
- Define a set of keywords for life satisfaction/people moods;
- Define a training set of data for machine learning people satisfaction.

7.1.2 Data access

Important potential restrictions are: the protection of personal data, copyright protection and public image protection. As no personal data will be gathered these restrictions will not apply.

On the other hand, regulations do not keep up with the development of new technologies. Potential problems may also include lack of agreement by the data owner (accessibility) and bureaucratic issues.

During the Web Meeting the participants concluded that one of the useful sources for Big Data is Twitter. Twitter can be used as a source of data for social science research both current and historical in-of-itself, but it can also be used to compliment more traditional data sources such as surveys and interviews. Taken into account were portals and websites, e.g. Facebook (regarding sentiment analysis)[5], Twitter (to find attributes on happiness)[6], Instagram, Google Trends. Data quality was assessed by rating selected dimensions of the data quality, e.g.: complexity, completeness, usability, timeliness and frequency, accuracy, coherence, validity, accessibility, clarity, etc. These aspects let us decide that Twitter is a valuable data source.

Although Twitter has less users, than Facebook which has about 3.3 million users, it has had a significant growth rate in recent years. The research of Gemius company shows that Twitter users 55% are men. The service is mostly used by people aged 25-34. However, according to Gemius there is a noticeably increasing popularity of Twitter in the 55+ age group. The service is most frequently used by journalists, publicists, heads of corporations, professionals, but also pupils and students; with a predominance of the latter. These groups constitute 60% of all users. The data also shows that Twitter is a platform that is being used by educated people. Mainly users of twitter live in cities (over 200 thousand residents). However the number of users from countryside is rising.

There are specific challenges associated with the use of social media data and in particular Twitter data. These challenges include ethical, legal and methodological issues.

Challenges on the use of social media data are as follows:

- Ethical issues: In collecting and retrieving data to form large datasets, it may not be possible to obtain informed consent from all of the participants. This is simply due to the volume of tweets retrieved. While the use of this data may be legal, it may not be ethical since the users providing the data may not have given informed consent.
- Legal issues: Sharing datasets is prohibited under Twitter's API Terms of Service. However, researchers can share the tweet identification numbers, associated with each tweet, which can be used by other researchers to obtain Twitter datasets.
- Retrieving datasets: Use of certain keywords or hashtags may not retrieve all of the data related to a topic. It may help if when brainstorming search queries as many queries as

[5] Ortigosa A., Martín J.M., Carro R.M., Sentiment analysis in Facebook and its application to e-learning; https://pdfs.semanticscholar.org/ffe0/0353d72de7cce64bcb797083b0ed39597ce6.pdf
[6] Quercia D., Ellis J., Capra L., Crowcroft J., Tracking "Gross Community Happiness" from Tweets; http://128.232.0.20/research/srg/netos/papers/p965-quercia.pdf

feasible are selected, and the dataset is filtered for non-relevant keywords after data-retrieval. This is because missing certain keywords or hashtags could introduce a systematic bias which would lead to a biased sample.

- Representation: Twitter users are not representative of the national offline population. As a matter of fact, Twitter users are not even representative among Internet users and Twitter data is not representative of Twitter users. This is because not all Twitter users will tweet on a topic of interest. For example, during the Ebola epidemic of last year not all Twitter users would post a tweet related to Ebola. It is also important to remember, it is not always individuals that may be tweeting but also organizations in a non-personal capacity, for instance journalists. Moreover, as the Demos and Ipos MORI report notes, the data that Twitter produces does not reflect Twitter users, as often a small number of vocal users account for a significant proportion of any given dataset[7].

Selecting the appropriate data sources is essential to the overall project success as it determines the choice and form of the pilot use case. Equally important is the recognition and analysis of Big Data sources, along with their content and availability, to facilitate determining the possible ways of using them in official statistics.

As the initial step, a brainstorming activity was undertaken among the project partners, as a result of which over 80 potential data sources for all domains were proposed:

Poland:

- **Human-sourced information (Social Networks): 16 ideas**
  - Social Networks: Facebook, Twitter, Tumblr etc. (3)[8]
  - Blogs and comments (1)
  - Pictures: Instagram, Flickr, Picasa etc. (2)
  - Videos: Youtube etc. (1)
  - Internet searches (4)
  - Mobile data content: text messages (1)
  - E-Mail (1)
  - Others (3)
- **Process-mediated data (Traditional Business systems and Websites): 28 ideas**
  - Data produced by Public Agencies: 14
    - Medical records (1)

7 http://blogs.lse.ac.uk/impactofsocialsciences/2015/09/28/challenges-of-using-Twitter-as-a-data-source-resources/

8 (x) – number of ideas

        - Others (13)
    - Data produced by businesses: 14
        - Commercial transactions (5)
        - Banking/stock records (1)
        - E-commerce (6)
        - Credit cards (2)
- **Machine-generated data (Automated Systems): 37 ideas**
    - Data from sensors: 29
        - Fixed sensors: **15**
            - ✓ Home automation (2)
            - ✓ Weather/pollution sensors (2)
            - ✓ Traffic sensors/webcam (6)
            - ✓ Scientific sensors (1)
            - ✓ Security/surveillance videos/images (3)
            - ✓ Others (1)
        - Mobile sensors (tracking): 13
            - ✓ Person (Mobile phone location) (5)
            - ✓ Road (Car, Trucks) (4)
            - ✓ Air (planes) (2)
            - ✓ Nautical (Ships) (2)
        - Satellite data: 1
    - Data from computer systems: 8
        - Logs (3)
        - Web logs (5)

Ireland:

- Mobile Phone Data
- Smart Meters
- Twitter
- Satellite Imagery
- Wikipedia
- Hotel websites
- Flight websites
- Traffic loops
- AIS Data

- Credit card data
- ATM Data
- Google Trends

Netherlands:

- AIS data
  - Movement of ships on Dutch waterways
- Call detail records –
  - Foreign visitors
  - Daytime population statistics (movement)
  - Visitors at festivals
- Google trends
  - Health statistics
- Road sensor data
  - Traffic intensity statistics
  - Economic indicators
- Scanner data
  - Price information based on scanner data
- Smart city data
  - Mobility of people during a festival
- Social media
  - Social media sentiment Indicator
  - Social media usage
- Twitter
  - Twitter topics
  - Social cohesion
  - Basic emotions
  - Safety feeling indicator
  - Dutch tourists
  - Foreign tourists
  - Detect seasonal changes
- Web pages
  - Price information from web pages
  - Job vacancies
  - Housing prices

    - Metadata of companies
    - Detection and estimation of animal population densities

UK:

- Job portals
    - Job vacancy statistics
    - Maintaining occupation classifications
    - Supplementary data on pay rates
- Enterprise websites
    - Job vacancy statistics
- Web scraped price data
    - Price quotes for CPI
- Retail Scanner data
    - Price quotes for CPI
- Google Trends
    - International migration
- Twitter
    - Internal migration
    - Sentiment analysis
    - Tourist sites visited
- Flickr
    - Tourist sites visited
- Electricity smart meters
    - Identifying vacant properties
    - Household type
- Travel smartcards
    - Commuting Statistics
- Mobile phone data
    - Commuting Statistics
- Sharing economy websites
    - Measuring impact of the sharing economy
- Credit card transactions
    - Locations of expenditure made abroad
    - Level of expenditure (and type?) made abroad

Afterwards, the project partners were requested to complete an expert questionnaire. As a result of these questionnaires the following low-availability data sources were identified:

- Mobile sensors (tracking) – Mobile phone location;
- Social Networks;
- Data produced by Public Agencies;
- Internet searches;
- Websites;

Both the brainstorming participants and experts completing the questionnaire indicated a number of major issues influencing the availability of data sources, which should be taken into consideration. These included:

- Legal aspects
- Data quality
- Methodology
- Technical aspects

Remarks concerning data quality, methodology and technical aspects can be found in Sections 7.1.3 – 7.1.5 of the Report.

Individual entities may refer to legal aspects such as copy right or image protection, while refusing to provide certain data. The refusal~~s~~ to provide data, as well as prolonged bureaucratic procedures, are likely to occur even if no legal limitations seem to apply. Therefore, based on experts' experience and responses provided in the questionnaire, certain data sources (e.g. tele-communication operators; databases) can be *a priori* considered unavailable.

It is worth noting, that Poland has no legal regulations determining the principles of the official statistics' access to automated measures (the Internet of Things). Laying down the principles of direct access to data coming from such sources would therefore clear any legal doubts which data owners and official statistics operators might have.

However, the lack of legal constraints should not imply that there are no other arguments against the use of certain Big Data sources. Special attention should be paid to ethical issues and social perception, in view of the growing social sensitivity as to the type of micro-data which can be gathered and used, and the methods employed for this purpose. For this reason, some data sources, e.g., databases maintained by tele-communication operators, seem to involve higher risks than generally-available data, e.g., the open Internet or social networking sites.

It is advisable to concentrate the actual resources and efforts on the sources which are considered easily available and do not contain personal data but, at the same time, include contents that might contribute to meeting the survey objectives.

7.1.3 Methodology

A common methodological framework has been determined for all WP 7 domains:

- No data sampling will be performed.
- A certain time frame will be determined for all data, depending on the survey needs (e.g. a day or a week).
- A common set of metadata concerning individual units (entities) will be established, its values being adjusted to the actual source-dependent needs.
- Data from unstructured sources will be analysed and adjusted to the key words selected before the survey.
- Text mining and machine learning tools can be used to analyse unstructured data.

The information separation methodology will be based on combining the records concerning a single entity. An entity forming part of the surveyed population can comprise a profile or characteristics of a person. This includes such attributes as location, gender, an intention to vote and mental wellbeing.

The available data sources for the population selected are generally unstructured. Differences also occur in terms of attribute sets and the possibility to aggregate several records for a given entity, depending on the data source. In the case of some sources, such attributes as age or location will be available, whereas in other cases they will have to be estimated by means of the adequate algorithms. This will influence the quality of results obtained from individual sources. The adequate application of weights in respect of the results obtained may solve the problems referred to above.

Lexical analysis and machine learning methods can be employed with the aim of determining the survey-specific attributes such as an intention to vote and the subjective mental wellbeing indicator. Literature sources provide a number of ready-to-use sentiment analysis[9] algorithms which could be potentially applied in the pilot use case.

The algorithm selection depends, among others, on:

- the planned lexical analysis level:
    - analysing the emotional charge of individual words, or words set in the discourse context,
    - considering the meaning of emoticons and (#)tags,

[9] Virgillito A., Vale S., UNECE Experiment report: Social Media- Sentiment Analysis, http://www1.unece.org/stat/platform/display/bigdata/Experiment+report%3A++Social+Media+-+Sentiment+Analysis

    - the number of emotional states identified,
- the efficiency of algorithms conditional on:
    - the data source type,
    - the source language
        - national language,
        - the use of a dialect/slang,
        - grammatical and spelling correctness,
    - the adequate preparation of glossaries and learner collections:
        - ampleness (the number of key words and the collection size),
        - the choice of the appropriate key words and their emotional charge,
- technical aspects of algorithm application:
    - the technical infrastructure and software required,
    - the skills required.

The algorithms to be used in the pilot use case will be selected at a later stage.

The survey stages can be described as follows:

1. Data acquisition – obtaining access to a given data source;
2. Data processing – as in the case of the ETL data warehouse – separating data from a given source by applying the rules mentioned above; transforming data into an integrated data set and uploading to the HDFS repository for the purpose of further analyses;
3. Data analysis – analysing data by means of statistical methods and generating a set of output tables and figures.

7.1.4 Technical Aspects

Our survey requires large quantities of data to be collected from the Twitter website and, at the subsequent project stage, also from other online portals. From the technical point of view, permanent broad-band access to the Internet and a dedicated working station constitute the only indispensable resources. Having accessed the Twitter websites (e.g. using the Google search engine) containing the intended topics/key words, user tweets (i.e. statements and comments) can be copied manually. A user account on Twitter is not required for that purpose.

To respond to these issues, Twitter provides a generally-accessible API (Application Programming Interface – a set of clearly defined methods of communication between various

software components […].An API specification can take many forms, but often includes specifications for routines, data structures, object classes, variables, or remote calls[10]). However, in order to use this function, creating an ordinary user account is required. Twitter's REST API enables making a maximum of 180 user-authorised requests or 450 application-authorised requests within 15 minutes, and the search is based on a strictly defined request[11]. Given the use of the Python-based development environment, the Tweepy library will be applied to implement the Twitter API within that platform. The machine learning algorithms will be activated in the Apache Spark environment using the Scikit-learn library.

In the case of other sources being web-scrapped, attention should be paid to certain restrictions on certain online portals. Using the generally-acceptable one-second interval between individual requests constitutes a good practice to avoid request blocking. It is also advisable to "introduce oneself" to the scrapping application by adequately setting the "user-agent" field. Numerous websites strictly indicate which paths can (or cannot) be accessed by search engine robots. Such restrictions are recorded in the robots.txt file located in the main service catalogue (e.g.). Nevertheless, such limitations seem to reflect partly technical, ethical and legal requirements, rather than constituting strictly technical aspects.

7.1.5 Quality issues

Availability is the most important data use criterion[12]. However, data use should be preceded by quality examination which also forms an essential element[13]. The quality of data to be obtained for the population in question can be generally considered average.

Firstly, the population surveyed will be limited to active users (i.e. users posting their comments) of the websites and social networking sites which constitute the survey data sources. As such, it may prove non-representative by excluding or over-representing certain social groups, depending on the source specificity. Some people using a given site are mainly inactive, while others do not use it at all. Data will therefore reflect only the attitudes of active users of social networking sites at a given time. The respondent group will be subject to continuous changes and comparing the

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[10] https://pl.wikipedia.org/wiki/Application_Programming_Interface, access 12.01.2017 r.
[11] https://dev.Twitter.com/rest/reference/get/search/tweets
[12] Hazen B., Boone C., Ezell J., Jones-Farmer L.A., Data quality for data science, predictive analytics, and big data in supply chain management: An introduction to the problem and suggestions for research and applications. [in:] International Journal of Production Economics. Aug2014, Vol. 154, 2014, pp. 72-80.
[13] Kwon O., Lee N., Shin B., Data quality management, data usage experience and acquisition intention of big data analytics. [in:] International Journal of Information Management. Jun2014, Vol. 34 Issue 3, pp. 387-394.

results of the survey, in the case of its subsequent repetition, will prove impossible. On the other hand, the cohort will change over time, but some active users will be able to be tracked over longer timeframes, so it might be possible to link a longitudinal cohort.

Secondly, the clarity of statements/comments may not be so obvious. It is of utmost importance to develop a glossary and to adopt precise definitions. Moreover, the fact that some statements can be ironic must not be neglected. The survey results obtained will also depend on the actual algorithm and learning patterns. Erroneous inclusions or omissions will therefore be unavoidable. However, according to human behaviour conception, negative comments tend to be more frequent than the positive ones. On the other hand, it becomes increasingly more common to post comments which do not reflect personal opinions but constitute a marketing attempt at eliciting specific reactions.

Thirdly, as regards the data sources which do not offer access to information included in the user profile, or do not require logging in, it is impossible to determine such attributes as age. Moreover, other attributes, like gender or location, may require to be estimated by means of certain algorithms, thus being subject to biases. What is more, the websites that do not require logging in provide limited opportunities, or make it entirely impossible, to identify, and then aggregate, a number of comments posted by one and the same user. In this event, user repetition errors and falsified results (resulting from a multiple analysis of data on one user, wrongly treated as several separate users) are likely to occur.

7.1.6 Use case

7.1.6.1 Preparation

Population domain mainly covers the study of analysis of the population attributes. The surveys' results are often presented as a population structure, i.e. the number of female per 100 men, as well as an index, i.e. net migration per 1000 population. Due to unrepresentative nature of the data present on the Internet and the possibility to apply the Big Data survey only on persons that are active in the Internet, it was decided to link the indicators of the population with the results of social surveys, such as life satisfaction or participation in the elections (intention to vote). The decision to choose this field of study is also the consequence of the possibility of comparing the results obtained by Big Data tools with the results from the traditional survey. The Eurostat on its website has the following subject groups:

- Looking for information on a specific topic
- General and regional statistics
- Economy and finance

- **Population and social conditions**
    - Population (Demography, Migration and Projections)
    - Population and Housing Census
    - Asylum and managed migration
    - Health
    - Education and training
    - Labour market (including Labor Force Survey (LFS))
    - Income, social inclusion and living conditions
    - Social protection
    - Household Budget Surveys
    - Youth
    - Culture
    - Sport
    - Crime and criminal justice
    - Quality of life indicators
    - Migrant integration
    - Equality (age and gender)
- Industry, trade and services
- Agriculture and fisheries
- International trade
- Transport
- Environment and energy
- Science, technology, digital society

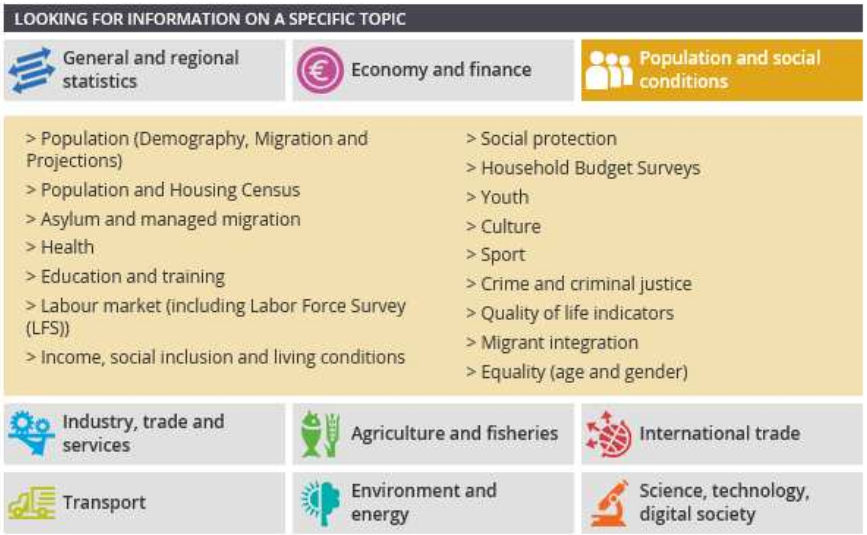

Fig. 1. The part of Eurostat's page concerning the selection of topics[14].

As can be seen in the list above, the proposed Use Case has been placed within the topics group 'Population and social conditions', and within the subgroup 'Quality of life indicators'.

It was decided to divide the use case into three parts:

1.1. Daily satisfaction (Life satisfaction)

1.2. The mood of population associated with public events (e.g., Brexit, Voting)

1.3. The morbidity areas (e.g., flu)

Accordingly, the aim of this study is:

1.1. To examine the level of daily satisfaction of the population by analyzing the content of messages for the presence of defined expressions describing emotional states, e.g., happiness, joy, sadness, fear, anger.

1.2. To present the moods of the population associated with various public events.

1.3. To observe morbidity areas, e.g., flu.

Population indicators will be limited to:

- Residence population (and migration);
- Number of women per 100 men;
- Population structure;

of persons using new technologies to communicate, such as social media, blogs etc. These indicators will be related to social statistics, such as life satisfaction, moods and morbidity areas.

Data obtained through the proposed solutions enable:

- Extending the scope of the database,
- Obtaining current results,
- Adding more detailed cross-sections for the study population of social media users and the Internet (currently there are no such sub-populations in similar studies).

In the first stage (December 2016 - February 2017) it is decided to conduct a pilot Use Case 1, e.g. Daily satisfaction within Life satisfaction. Other Use Cases in Population domain, number 2 and 3 will be made in the period of February and March 2017.

Tab. 1. *Schedule for carrying Use Case*

Use Case	12.2016	01.2017	02.2017	03.2017
1.1.				

[14] http://ec.europa.eu/eurostat

1.2.				
1.3.				

The criteria for selecting the Use Case is primarily related to the availability of the data source and the reliability of the results. Therefore it was decided to use the widely available source, access to which is possible by using web scraping or a dedicated API.

It was assumed that the life satisfaction data may be based on the data sources such as social media and web pages. The advantage of social networking sites in relation to websites is the possibility of obtaining the geographical location of the entry[15]. Concretizing the data source, the following portals / tools were indicated:

- DS1 – Twitter
- DS2 – Google Trends
- DS3 – Comments on Specific News/Events on Web Portals such as gazeta.pl, bbc.co.uk, irishtimes.com, spiegel.de

According to the data quality assessment (point. 7.1.5), these sources are characterized by a relatively high quality of data, that is sufficient to be acquired and processed in order to achieve the main objective of the Use Case.

7.1.6.2 Description

The main aim of Use Case 1.1 is to obtain data on Daily Satisfaction. This means that satisfaction of life is estimated by analyzing the entries posted on the Internet. To achieve this objective, it was decided to use the API tools to retrieve data from the social media portal Twitter.

Examples of key words and questions related to the study of life satisfaction from the questionnaire European Social Survey were presented in Figure 2.

[15] Schwartz H.A. et al., Characterizing Geographic Variation in Well-Being Using Tweets; http://www.aaai.org/ocs/index.php/ICWSM/ICWSM13/paper/viewFile/6138Hansen/6398

		None or almost none of the time	Some of the time	Most of the time	All or almost all of the time	(Don't know)
D5	…you felt depressed?	1	2	3	4	8
D6	…you felt that everything you did was an effort?	1	2	3	4	8
D7	…your sleep was restless?	1	2	3	4	8
D8	…you were happy?	1	2	3	4	8
D9	…you felt lonely?	1	2	3	4	8
D10	…you enjoyed life?	1	2	3	4	8
D11	…you felt sad?	1	2	3	4	8
D12	…you could not get going[31]?	1	2	3	4	8

Fig. 2. Example of questions in the European Social Survey questionnaire as the framework of questions about feelings from last week

Due to the nature of entries on social networking sites it is not possible to acquire such detailed data on life satisfaction as it is, e.g., in the European EU-SILC (European Union Statistics on Income and Living Conditions) within the module EU 2013 - Wellbeing (Fig. 3).

Average rating of satisfaction by domain, sex, age and educational attainment level **[ilc_pw01**

Last update: 15-09-2016

Table Customization show

INDIC_WB | GEO | Unit of measure: Rating (0-10)

International Standard Classification of Education (ISCED 2011): All ISCED 2011 levels | Period of time (a=annual, q=quarterly, m=monthly, d=daily, c=cumulated from January): 2013 | Sex: Total

Age class: 16 years or over

INDIC_WB / GEO	Satisfaction with fin	Satisfaction with acc	Job satisfaction	Satisfaction with cor	Satisfaction with tin	Overall life satisfact	Satisfaction with rec	Satisfaction with li
European Union (28 countries)	6.0	7.5	7.1	7.4	6.7			7.
Belgium	6.9	7.8	7.5	7.6	7.1	7.6	7.3	7.
Bulgaria	3.7	6.0	6.0	5.9	5.7	4.8	5.2	5.
Czech Republic	6.0	7.7	7.4	7.5	6.7	6.9	7.5	7.
Denmark	7.6	8.4	8.1	8.3	7.8	8.0	8.3	8.
Germany (until 1990 former t	6.3	7.5	6.9	7.5	6.5	7.3	7.6	7.
Estonia	5.4	7.1	7.3	7.4	6.7	6.5	7.0	6.
Ireland	5.5	8.0	7.2	8.0	6.9	7.4	7.5	8.
Greece	4.3	6.6	6.1	6.5	6.1	6.2	5.9	6.
Spain	5.8	7.3	6.9	7.1	6.6	6.9	6.6	7.
France	6.4	7.5	7.2	7.5	6.9	7.1	7.2	7.
Croatia	4.6	6.9	7.0	7.2	6.6	6.3	5.8	6.
Italy	5.7	7.2	7.0	7.0	6.4	6.7	6.1	6.
Cyprus	5.2	7.7	7.2	7.3	6.7	6.2	5.9	6.
Latvia	5.0	6.6	7.3	7.2	7.1	6.5	7.5	7.
Lithuania	5.8	7.4	7.5	7.7	6.8	6.7	7.4	7.
Luxembourg	6.9	7.8	7.5	7.7	7.2	7.5	7.8	7.
Hungary	5.2	6.8	7.1	7.1	6.3	6.1	6.2	6.
Malta	6.0	7.9	7.5	6.6	6.6	7.1	6.6	7.
Netherlands	7.4	8.1	7.7	8.0	7.5	7.8	8.1	8.
Austria	7.0	8.3	8.0	8.0	7.3	7.8	8.3	8.
Poland	5.7	7.4	7.3	7.5	6.8	7.3	7.5	7.
Portugal	4.5	7.3	7.0	7.5	6.5	6.2	6.0	6.
Romania	6.2	7.4	7.0	7.1	6.8	7.1	7.0	7.
Slovenia	5.6	7.6	7.3	7.8	6.8	7.0	7.9	7.
Slovakia	5.5	7.6	7.2	7.3	6.9	7.0	6.8	6.
Finland	7.5	8.4	8.1	8.2	7.7	8.0	8.3	7.

LIFESAT - Overall life satisfaction

Fig. 3. Example of indicators of wellbeing Eu2013 module for EU-SILC

The main advantage of the proposed solution is the possibility to carry out surveys more frequently comparing to the currently survey's modules that are usually conducted every few years. In addition, the benefit is this method is not a burden on respondents.

The solution involves the use of stages according to the process presented in Figure 4.

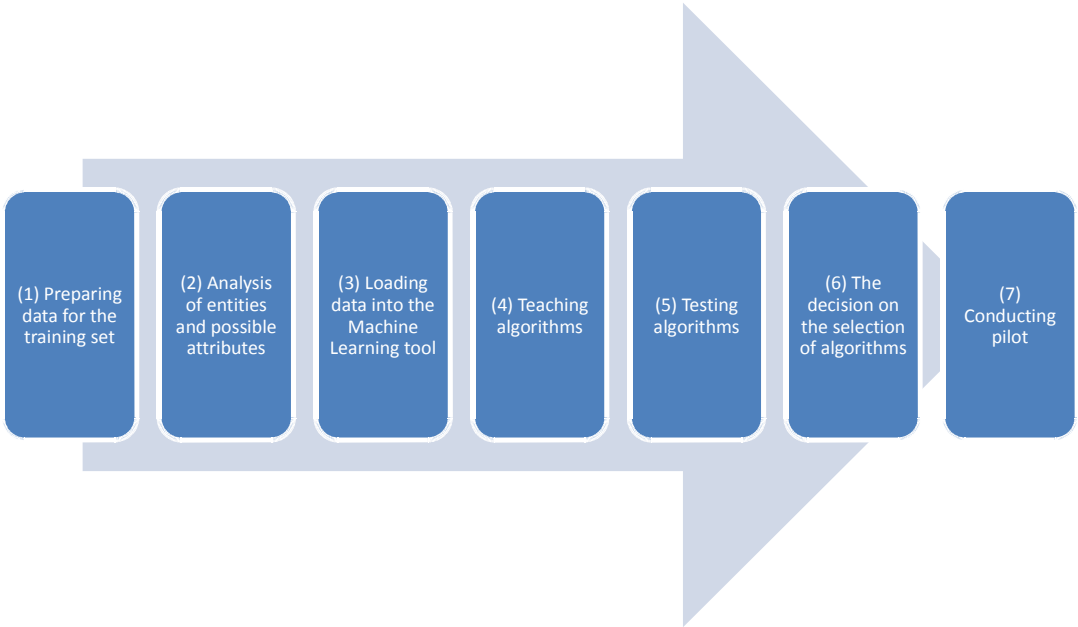

Fig. 4. The process of implementation of the Use Case

According to Figure 4, Use Case 1.1 realization includes 7 steps. Within these stages the following tasks will be realized respectively:

- Classification of text obtained in order to extract the attributes related to life satisfaction.
- Analyzing possible attributes of the population to obtain.
- Preparing Python solution to enable the acquisition of tweets and processing them with algorithms related to Machine Learning methods.
- Machine Learning algorithms training to find the type of text classification.
- Verification of the algorithms on the training data set.
- The selection of algorithms and data sources to conduct the survey in a test environment.
- Conducting a pre-pilot in a production environment.

In SGA-1 we decided to test this environment in pre-pilot conditions. The pilot survey will be conducted in SGA-2.

More precisely it must be noted that the classification referred to the first task includes the analysis of emotional states according to a survey EU-SILC, i.e.:

• Very upset;

• So deeply depressed that nothing can lift your spirits;

• Quiet and calm;

• Discouraged, nailed or had the blues;

• Lucky.

Within the first phase, the solution uses Python version 3, and the environment Apache Spark for data acquisition and processing. For the implementation issues related to the webscraping, Twitter API library Tweepy was used. In order to implement the Machine Learning algorithms, scikit-learn library was used. The whole framework is presented in Figure 5.

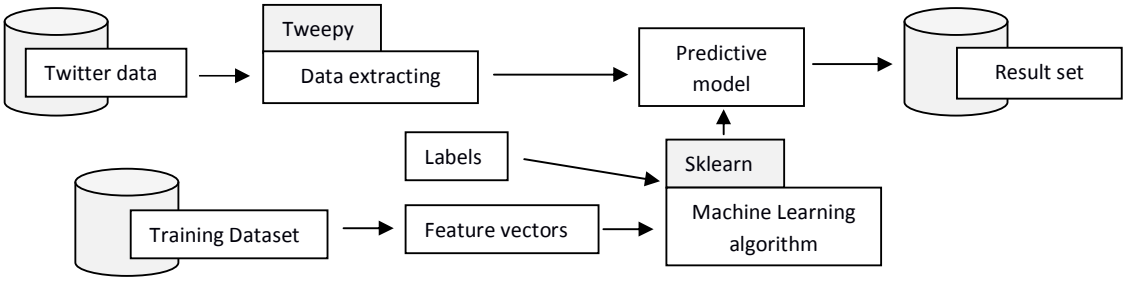

Fig. 5. *Big Data Framework used for Use Case 1.1.*

Training set looks as presented in the Table 2 examples.

Tab. 2. The content of the training set

No.	Text	Target	Language	Id
1	Rousey is gonna quit UFC forever now lmao #SoHappy	Satisfied	EN	F1
2	And I did absolutely nothing #satisfied 😄🎉💕	Satisfied	EN	F2
3	To był cudowny weekend ☺😇💝💗💓 #love #happy #awesome #osom #bestweekend @ Czestochowa	Satisfied	PL	F3
4	Połączenie nowoczesnego designu z funkcjonalnością sprawi, że osiągniesz jeszcze lepsze wyniki.	Neutral	PL	F4
5	They want more happiness & more money in 2017 cause they're not satisfied w/the position of each. It don't matter the context. #Unsatisfied	Not satisfied	EN	F5

In summary, the general outline of this study includes the following topics:

- Webscraping,
- Analyzing of comments / messages,
- The selection and arrangement of useful information,
- Quantitative and qualitative classification of posts/messages/comments (machine learning).

Before the analysis the weaknesses of the proposed solution must be taken into account. In includes the following:

a) Data from social networks are information about the state, they cannot predict developments;
b) Not representative – the data will only reflect the mood of people who are active in social networks;
c) Data from social networks are generally unstructured and appear irregularly.

To sum up, we can say that the project sets up the use of large sets of data from social networks to assess the current state of the moods of people associated with the public event as well as related indicators, such as life satisfaction or studying the impact of public mood on the decision. Particularly interesting could be the analysis of the moods of citizens of different countries in relation to the same events and mood changes over time.

The work plan for the future is to develop more indicators associated with satisfaction in various areas of life, e.g., work, commuting time etc. A multitude of these indicators is presented in Figure 2.

7.1.7 Combining data

Based on the methodological assumptions regarding the Use Case, the data combination process is to be conducted according to the following three scenarios:

1) Integrating data from various data sources related to a given topic – e.g.
   1.1. Daily Satisfaction;
2) Supplementing the survey results obtained through traditional questionnaires by adding new and more detailed results to statistical tables, assuming that the results obtained are not representative (referring to a certain sample);
3) Adding new data to the output tables compiled for traditional surveys, making use of the Big Data tool.

According to the first data integration scenario, the data combination process will be conducted within complementary Big Data sources. For instance, Twitter data is planned to be combined with data coming from the comments posted on various websites. Such a process will

require separating certain attributes from the entity described by means of the text mining algorithms. These attributes will refer to:

1) the topic
2) the sentiment (attributes concerning, among others, life satisfaction)
3) gender
4) geographic location
5) other attributes available.

Assuming the lack of an obvious gender and geographic location indications in typical news portals, the data combination model in question requires selecting the data sources for which at least the first two of the attributes mentioned could be available. In this event, these data sources will be combined based on the attribute values.

In view of the varying representativeness of data sources available on the Internet, the data combination process is planned to cover the sources displaying similar representativeness levels.

As regards the second data combination scenario, which provides for supplementing the results obtained from traditional surveys, the aim is to extend the scope of the currently-available results by making use of Big Data sources. This requires selecting such Big Data sets that perfectly correspond to the objectives assumed in traditional surveys. For methodological reasons, not all traditional surveys can be supplemented with data coming from Big Data sets. The number of such surveys is, in fact, very limited and concerns mainly various population features. Given that the reference sample is restricted to Internet users, it is assumed that the population surveyed will correspond to that included, among others, in ICT surveys.

The third scenario envisages adding more detailed information to the currently-available data by including additional cross-sections. As in the case of the second scenario, the population is planned to be limited to active users of social networking sites. The sample based on that population could be described in further detail.

Summing up, there are three general methods of combining data sets, which were specified in the WP 7 description:

- Combine in one repository the selected data from all Big Data sources,
- Supplement of information gained in traditional surveys,
- Comparison with the results of traditional surveys to add more detailed information.

7.1.8 Summary

Population is one of the few domains that has been subjected to statistical surveys in many areas. Data published by the internet user allows the diagnosis of social moods, the moods of people

associated with various public events, or to observe the impact of public moods on the decision but also we can observe morbidity areas, e.g., flu or examine the level of daily satisfaction of the population by analyzing the content of messages for the presence of defined expressions describing emotional states, e.g., happiness, joy, sadness, fear, anger.

There are specific challenges to using social media data including ethical, legal and methodological issues. Official statistics have value as they can support and enrich traditional public opinion research e.g: Support traditional European Social Survey, supplement of the research methodology of some phenomena that are difficult to measure through traditional polls.

Deliverable 7.2. Report on *Tourism/Border crossing* domain area containing basic information on the data access, quality issues, methodology (focus on combining data) and technical aspects.

7.2.1. Background

Tourism is a complex and multifaceted phenomenon, which refers to many aspects of human life, both in terms of social and economic. It occupies an important place in the economy of the EU Members because it contributes to employment, economic growth, development and socioeconomic integration in rural, peripheral or underdeveloped areas. To assess its competitiveness, good statistics are acquired on the volume of tourism, its characteristics, tourism expenditure and the share of tourism within European economy. The economy-wide effects of tourism are significant and vary a lot from one territory to another in terms of number of visitors, type of tourism, seasonality patterns, revenues and added value per visitor or jobs generated.

Thanks to tourism, people regenerate physical and mental strength, discover the world and form their own personality. Tourism is also a form of economic activity, within which developed various kinds of tourist services offered to travelers out of which the most important are: accommodation services, catering services and transport services. Tourism statistics of XXI century is constantly evolving, making the improvements in data collection a necessity. The needs are focused on providing reliable and comparable data that take into account the budget constraints. Providing speedy information which can allow tourism organizations to react instantly is also absolutely crucial. Innovations in the field of collecting and compiling tourism statistics meet increasing user needs to reduce the burden on respondents and administration, as well as cost reduction. Changes related to the technological and digital revolution in recent decades opened up interesting prospects. Wide access to the use of mobile phones, GPS devices or wireless internet means that the traveler leaves a "digital footprint", which is a potential source for the analysis of tourism statistics. Information obtained from mobile network operators, from booking or reservation systems, from internet activities, from electronic payment cards, or from social media can be used by producers of statistics. At the moment, while it is difficult to predict the future of Big Data, the potential of tourist organizations working with the information generated arguably will be of value, and may bring an improvement in the customer experience.

Some potential new data sources appeared but political changes connected to establishing Schengen Area resulted in loss of information about border traffic. As a result of the inclusion in the Schengen Information System, the Border Guard discontinued the registration of movement of persons and means of transport at the borders with the countries of the European Union. New

regulations on customs clearances resulted in the loss of data on border traffic at the borders with EU member states.

Each country tried to path its own way to deal with that problem by using other sources of information. Therefore, methodological work has been undertaken in order to develop methods for collecting the missing data on border traffic. One of them is to conduct sample survey. Border traffic is essential to secure the information needs of official statistics in the tourism area, balance of payments, national accounts and cross-border areas.

However, it turns out that more complete and detailed data on border traffic can be obtained from Big Data sources such as traffic sensors located in relevant placed along the border. These data are currently available for Road and Motorway Directorate in many countries. Implementation of these data in a traffic estimation process would significantly decrease burden of interviewers and the need for manual vehicle counting as well as cost of the survey. Moreover, it may be used to produce some estimations and projections with higher frequency.

7.2.2. Data access (with legal and ethical issues)

With the development of new technologies that offer huge data sets such as Big Data are growing concerns about the border crossing of privacy and use of personal data for commercial purposes. The use of Big Data available through the generation of information from social networking sites or use to enter the network of mobile devices raises questions about the consequences in terms of the privacy.

Project participants analyzed the potential sources of data for the possible development of statistics in the domain of tourism / border traffic. Among others the following potential data sources were mentioned:

- AIS (automatic identification system) data,
- Road sensor data,
- Smart city data,
- Twitter, Facebook or other social media channels,
- ATM (asynchronous transfer mode) Data,
- Credit card data,
- Flickr,
- Flight websites,
- Google Trends,
- Hotel websites,

- Mobile phone data,
- Traffic loops,
- Travel smartcards,
- Wikipedia.

Taking into account the availability of data from the road traffic sensors the members of the SGA-1 project decided to use this source for conducting a pilot survey. No legal problem should result from the exploitation of Big Data generated by traffic sensors in Poland. Big Data are generated, through an opened competition, by the General Directorate of National Roads and Motorways (GDDKiA) for technical and planning needs. These data are used by public administration (e.g. national police, Central Statistical Office of Poland, etc.). Therefore, this Big Data source does not contain information on individual persons. This excludes the main source of legal problems related to privacy violation.

Tourism statistics offers a big opportunity for exploring new data sources and methods. However using sources other than statistical surveys as a source of information even partially will become a big challenge for statisticians. First of all access to the Big Data can may highlight some issues relating to ethics.

Part of the project 'will' be to observe–the following ethical aspects[16] that may be associated with the mentioned data sources for use in the field of tourism statistics and border traffic.

1. Access to the data - independence and impartiality

The Big Data collected by private companies e.g. mobile positioning data, scanner data, social media data etc. usually are not available for statistics for free. In order to get access to the data it is necessary to contact data providers respecting the principle of impartiality and independence of statistical authorities.

All the data providers have to be treated equally ensuring, for instance, equal opportunities to improve their market positions. When selecting the ways of collaboration (partnerships) with the Big Data holders the statistical authorities have to retain professional independence in producing and disseminating of official statistics.

2. Purchasing of the data - independence, professionalism

The question is if purchasing of Big Data from data providers is ethical for statistical authorities. Regular purchases of the data sources by statistical authorities could be seen as incompatible with statistical professional values. The Big Data might have a commercial value for the data provider, being in some cases a core of its business model. The main aim of commercial

[16] Based upon a document which was managed by SOGETI on the request of Eurostat.

relationship is to generate financial gains. The NSIs (and quality of the end product) may become dependent on available financial resources.

3. Trust in Big Data providers - responsibility

Statistical authorities have to be transparent on the statistical methods and procedures used but the collaboration with enterprises that are lacking transparency may compromise the reputation of official statistics. Cooperation with the Big Data providers should be under condition that they are compliant with national law and respect privacy of data subject.

4. Data subjects' consent - respect the privacy of data subjects

The use of Big Data might raise significant privacy concerns. For statistical authorities to work in ethical terms means respect the privacy of data subjects. The data subjects (individuals and companies) have to be aware that their data collected by other companies are used for the purposes of official statistics. It could be done by the Big Data holders. In order to respect the privacy of others the NSIs should have the data subjects' consent to use their sensitive private data (medical, financial etc.) obtained from the third parties for statistical purposes.

5. The data linking - confidentiality of individual data

Big Data used for data linking by NSIs may reveal personal data and raise an ethical issue. In order keep strong commitment to confidentiality of individual data all the Big Data sources that contains directly or indirectly identifiable personal data have to be securely protected. Any possibility to compromise individuals or organizations should be excluded. Strong rules of data security have to be implemented and communicated to general public.

6. Continuity and stability of Big Data sources - quality of data sources

Big Data are not designed for statistical purposes and do not meet statistical standards on concepts and definitions. If a Big Data holder decides for business reasons to change definitions, collect different data, or entirely stop their data collection, statistical authorities have virtually no leverage to prevent the loss of such data. In case if there are no appropriate safeguards to protect continuity and stability of the data sources it would be very risky to base official statistics on them.

7. Manipulation - quality of data sources and transparency

Data derived from third parties, for example social network data or voluntarily contributed data, bear the risk of being manipulated by the data provider itself or by third parties and it may be difficult to detect. To work in ethical terms for official statistics in this case mean to apply different quality checks to the data extracted from Big Data sources. If manipulation cannot be excluded or cannot be detected and corrected, a certain data source may not be usable for the purpose of producing official statistics.

8. **New skills /outsourcing Big Data analytics - professionalism**

Big Data analytics require special skills for analysis of a high volume of data from a variety of structured, semi-structured, unstructured or incomplete data. The level of complexity may require tools and models not used in official statistics. There is a risk for official statistics to lose professional independence when it is outsourced to Big Data analytics. The difference is that the analysis of the data is a core business of statistical agencies. In order to remain independent and develop professionalism the NSIs could form partnerships with companies that possess the required skills and tools with the role of statistical agencies as guarantor of quality and impartiality. Methods applied by the data analysts have to be transparent.

9. **Complexity of Big Data analysis - transparency**

Professional ethics requires that "Statistical agencies have to be transparent about the statistical methodologies used and make these methodologies public." In cases no clear methodology can be supplied, Big Data can only be used as auxiliary information to complement official statistics.

10. **Cooperation or competition**

This is due to the face that, there are other players in the information market that use Big Data for statistics. If statistical agencies compete with private companies it might be considered as a duplication of work and a waste of public resources.

NSIs should collaborate rather that compete with the private sector working with Big Data under the leading role of the statistical agencies that would increase potential of official statistics and save public resources.

7.2.3. Methodology

One of disadvantages of Internet-based Big Data sources is lack of methodology how to deal with such non-structuralized data. We observe following obstacles:

- proper defining surveyed population
- non-representative sample
- enlisting all websites relevant to task (defining survey frame)
- behaviour in Internet may be not similar to observed in real life

Therefore, Internet-based Big Data sources should be tackled with great caution and by experienced statistical units.

Mobile phone data seems to have less methodological problems compared to Internet-based Big Data sources. Nevertheless, mobile phone data owners state there may be some problems with identification of calls just near the border of Poland. It may happen that a person calling from Polish

territory, situated very close to Germany is recorded as a person calling from Germany because the Base Transceiver Station located in Germany is closer than that one in Poland. To sum it up, Polish borders are not identical to borders of nets of Polish Base Transceiver Stations.

Credit card data is considered to have a disadvantage in its lack of representativeness. Not all of tourists use credit cards. On the other hand, more and more transactions are done without cash nowadays, thus it might be very interesting and methodologically transparent Big Data source in close future.

Data from traffic sensors produced by public agency such as GDDKiA has got very clear methodology of collecting data, carrying out surveys including measurement errors. GDDKiA uses automatic, the semi-automatic sensors and the manual vehicle registration. The automatic sensors are very precise and are connected to the system ETC viaTOLL and without interruption registers data on the vehicle movement during the whole year. This system is used for the registration of vehicles on the motorways. The half automatic sensors (Golden River M660/M680, PAT AVC 100, RPP-2, RPP-3, RPP-5) are characterized by lower precision. It combines automatic and manual vehicle registration. The type of traffic sensor used depends on the specificity of the road segment, while the manual registration can be applied in all kinds of roads.

The timing for carrying out the traffic registration is identical for all types of vehicles (16 hours during the day, that is from 6h00 to 22h00 and 8 hours during the night time, that is from 22h 00 to 6h 00). For truck vehicles, special timing has been prepared during the year 2015.

Following the above considerations and those included in the documentation of the GDDKiA on methodology for the Polish traffic registration, one may conclude to its conformity with the EU system of weighting vehicles on the traffic.

However, the next potential problems can impact the quality of data as no methodology exists to deal with following issues:

- new regulations on toll charges which may change the vehicle traffic density on a given segment,
- extension of the number of toll charge points on a given segment,
- changes in traffic density consecutively to atmospheric conditions. Such changes should impact, on its own way, tourism movement also,
- changes in traffic density consecutively to administrative regulations or to random events, like road accidents.

One issue that arise is increasing number of traffic sensor. It would not be appropriate to omit new information in modelling. Model calculated at one moment of time may be outdated one year later due to new data available. It requires updating model definition and parameters each year.

The second issue it is worth to mention are changes in road network e.g. opening new segment of motorway. It is clearly visible in time series as a huge fall in traffic intensity within very short period of time for certain roads. Impact of changes in road network may be very hard to estimate if the change is made at the same moment as the moment of forecasting. Even if more data is available it requires specific approach e.g. modelling decrease of traffic with dummy variables.

7.2.4. Technical aspects

Technical aspects were assessed for several Big Data sources. Internet sources such as Twitter, Flickr, Google Trends require using programs exploring internet. At this moment we have little experience with such programs. Moreover, for example geolocated Twitter data processed by Office for National Statistics consisted of over 106 million of tweets[17] which required relevant hardware and software.

Similar challenge occurs with mobile phone data and credit card data. Thus, more aggregated data are more favorable.

Data obtained from public agencies such as General Directorate for National Roads and Motorways (GDDKiA) is already aggregated to relevant level. Thus, the database is of small volume. For instance, a database for 2015 contains around 5 thousand measurement points. In a case of estimating border traffic nearly 300 of measurement points are relevant to our task in 2015. The optimum result would be an estimation of traffic on all national roads. In this case, a whole data base is required. All the tests carried out already on that database were passed well. The problem that may arise is processing of distance matrix or adjacency matrix. It is not a case of estimation of border traffic because we shall use data only from 300 measurement points and that matrix will have 90 000 elements (300 times 300). It may be in case of estimation of traffic on all national roads because that matrix will have 25 million elements (5000 times 5000). Nevertheless, this problem may be tackled with methods of processing spare matrices since more than 99% elements of adjacency matrix would be zeroes.

[17] Swier N., Komarniczky B., Clapperton B., Using geolocated Twitter traces to infer residence and mobility. GSS Methodology Series No 41, 2015

7.2.5. Quality issues

Quality of Internet-based Big Data sources is rather poor. Experience gained by several statistical offices reveals that it requires a lot of efforts to derive valuable data from Internet-based sources. Previously mentioned methodological issues affect their quality also.

Credit card data is considered as a source of high quality due to domain and law restrictions.

Mobile phone data seems to be of a good quality up to methodological issues mentioned previously.

Data obtained from public agencies such as General Directorate for National Roads and Motorways (GDDKiA) is of open access. Public agencies generally produce data of a very high quality. GDDKiA traffic sensor Big Data are produced by this institution mainly for its own uses. Then, two questions may arise.

The first question considers at which extent Big Data initially conceived mainly for the own needs by the GDDKiA can fit to the need of tourism/cross border use case. The GDDKiA registers vehicle movement at Polish border and inside the country along national roads. Since a strong correlation potentially exists between touristic activity and vehicle movement, one may apply ad hoc filters on these Big Data - e.g. selecting a certain category of vehicles during a particular year period- for the touristic activity modelling.

Since we accept that the GDDKiA Big Data remain valuable for tourism modelling, the subsequent question concerns their quality. The GDDKiA uses two types of electronic sensors to generate, on a daily basis, Big Data obtained with the international quality standards. The first category of sensor is fully automatic and the error on the Big Data production is technically not greater than 1%. These kind of devices are used on motorways and information concerning each category of vehicle is generated. The second category of sensor is semi-automatic (with camera video) and related error on produced BD is not greater than 3%. Thus, this category combines automatic and manual vehicle registration. For this reason, outputs generated by this sensor are global and do not display details of each vehicle. Furthermore, to enhance the quality control of data generated by this sensor, a manual vehicle registration is parallelly carried out.

Furthermore, all traffic sensors are regularly tested by the Warsaw GDDKiA technical organ (DPI WAR GDDKiA) assisted by its Gdansk branch.

Thus, one can consider that the quality of produced data remains on the international standard level.

A factor that makes data concerning traffic hard to analyze are different labels of traffic sensors in databases in two surveys conducted by GDDKiA: (i) General Traffic Measurement and (ii) Continuous Traffic Measurement. Also there are minor shifts of location for some sensor and slightly

different estimates in a result. We dealt with location issue by mapping traffic sensors from General Measurement to the closest one in Continuous Measurement because traffic intensity was very similar in both measurements. On the other hand we kept values of both measurements since estimations error for General Measurement is lower than for Continuous Measurement and we want to put that information in a model.

7.2.6. Use case

We present that subject in two steps. The first one concerns how we made a path to the data sources selection and the use case while the second one activities already carried out as well further plans of data processing.

7.2.6.1. Toward use case

The first step taken to identify pros and cons of Big Data sources was brainstorming session conducted independently by group from Poland, Netherlands, United Kingdom and Ireland. Participants listed Big Data sources and assigned suitable use case and relevant domain. We selected those results which pertain to domain of tourism, border traffic and transportation. Domain of transportation was selected also because it may contribute to border traffic estimation since border traffic is a part of traffic in general.

Group from Netherlands indicated following Big Data sources useful for domains of interest:

- AIS data,
- Call detail records,
- Road sensor data,
- Smart city data,
- Twitter.

Call detail records were used to produce statistics on foreign visitors (tourists) to the Netherlands as well as to estimate the number of visitors to several festivals in the Netherlands. It seems to be very adequate and promising. There is often a need to estimate amount of tourists in general and also number of tourists which visited the country to participate in an event that is a subject of interest.

On the basis of road sensor data for Dutch highways, monthly/quarterly traffic intensities are calculated for each road per COROP region (regional area within the Netherlands.). The project results in official statistics. The next step that is already undertaken is the possibility to combine data

sources such as road sensors in motorways, GPS data and camera footage with information from questionnaires and subsequently to estimate how goods are transported over the Dutch road network. PhD student Yinyi Ma from Erasmus University investigated a theoretical model (a hierarchical Bayesian network) enabling her to combine data from various sources. The model was successfully tested but has not been used in practice[18].

Twitter data were explored to assess the selectivity of a Big Data source. CBS staff explored different ways of extracting auxiliary information, both from the Big Data source itself and from linking it with other sources of information. They applied profiling method to a dataset of Dutch Twitter users[19].

Some works were performed during the DataCamp. Project studied the feasibility of detecting foreign tourists and camping sites in the Netherlands based on Twitter data.

Group from United Kingdom indicated following Big Data sources useful for domains of interest:

- Flickr,
- Twitter,
- Travel smartcards,
- Credit card transactions,
- Mobile phone data.

ONS conducted a pilot study to investigate Twitter with respect to possibility to infer a user's residence and analyze mobility patterns. They found out that it is feasible to cluster geolocated activity traces with Density Based Spatial Clustering Algorithm with Noise from Twitter to infer a user's location of residence providing there is sufficient user data. However, they concluded that half of all geolocated tweets were made by just 4 per cent of users while 17 per cent of users only sent one tweet. Thus, the volume of geolocated Twitter activity is very uneven across the user base. Geolocated Twitter data is uneven and patchy across the user base. Collection of data from social media can be impacted by a combination of technological change (including those of third parties) and the behavioural response of users. This has implications for time series analysis and illustrates why caution is needed when using social media data to inform decision-making. Social media data is unstable and may be affected by unexpected technological and behavioural changes[20].

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[18]https://www.cbs.nl/en-gb/our-services/innovation/nieuwsberichten/recente-berichten/new-steps-in-big-data-for-traffic-and-transport-statistics

[9]Daas P.J.H. et al. Profiling of Twitter users: a Big Data selectivity study, Discusion Papers 06, 2016

[10] Swier N., Komarniczky B., Clapperton B., Using geolocated Twitter traces to infer residence and mobility. GSS Methodology Series No 41, 2015

Group from Ireland indicated following Big Data sources useful for domains of interest:

- Flight websites,
- Hotel websites,
- Wikipedia,
- Twitter,
- Google Trends,
- ATM Data,
- Credit card data,
- Traffic loops,
- Mobile Phone Data,
- AIS Data.

The second step we have taken was conduction of online survey on Big Data sources among 19 statistical institutions. Results of questionnaire pointed out pros and cons of data sources and allowed us to specify appropriate Big Data sources for our domain.

Table 1Table 1 shows which Big Data sources are relevant to tourism and border traffic domains.

Table 1 Big Data sources useful for tourism and border traffic

Big Data Sources		Useful for Tourism and Border Traffic	
		N	%
Internet searches		5	26
Websites		7	37
Blogs and comments		3	16
Data from computer systems – Logs		2	11
Data from computer systems - Web logs		2	11
Data from sensors - Fixed sensors -	Security/surveillance videos/images	1	5
	Home automation	0	0
	Scientific sensors	0	0
	Traffic sensors/webcam	4	21
	Weather/pollution sensors	1	5
Data produced by businesses - Banking/stock records		1	5
Data produced by businesses - Commercial transactions		3	16
Data produced by businesses - Credit cards		8	42
Data produced by businesses - E-commerce		1	5
Data produced by Public Agencies		6	32
E-Mail		2	11
Mobile data content: text messages		2	11
Mobile sensors (tracking) – Cars		4	21

Mobile sensors (tracking) - Mobile phone location	15	79
Mobile sensors (tracking) - Satellite images	0	0
Personal documents	1	5
Pictures: Instagram, Flickr, Picasa	1	5
Social Networks: Facebook, Twitter, Tumblr etc.	6	32
User-generated maps	2	11
Videos: Youtube etc.	1	5

Survey reveals that top Big Data sources are mobile phone location (79%), credit cards (42%), websites (37%), data produced by public agencies (32%), social networks (32%), internet searches (26%) and traffic sensors/webcam (21%). At first look we can divide them into two groups: mobile phone location, credit cards, data produced by public agencies and traffic sensors/webcam are sources with quite nice structure. Database contains full information pertaining to subject or it is a sample but there already exists a methodology of data imputation and estimation. The second group of sources is not of good structure. They all come from Internet, require more advanced methods of preparing database and processing. There is no database ready to use for them. Table 2 shows assessment of difficulty to access certain Big Data sources.

Table 2 Access to Big Data

Big Data sources	Access					
	Easy		Medium		Difficult	
	N	%	N	%	N	%
Internet searches	10	67	4	27	1	7
Websites	13	76	3	18	1	6
Blogs and comments	5	36	7	50	2	14
Data from computer systems - Logs	0	0	6	55	5	45
Data from computer systems - Web logs	1	9	6	55	4	36
Data from sensors - Security/surveillance videos/images	0	0	1	9	10	91
Data from sensors - Home automation	0	0	2	15	11	85
Data from sensors - Scientific sensors	0	0	5	45	6	55
Data from sensors - Traffic sensors/webcam	3	19	9	56	4	25
Data from sensors - Weather/pollution sensors	0	0	8	67	4	33
Data produced by businesses - Banking/stock records	0	0	2	15	11	85
Data produced by businesses - Commercial transactions	0	0	4	29	10	71
Data produced by businesses - Credit cards	0	0	4	27	11	73
Data produced by businesses - E-commerce	0	0	5	38	8	62
Data produced by Public Agencies	8	57	5	36	1	7
E-Mail	0	0	2	15	11	85
Mobile data content: text messages	0	0	1	8	11	92
Mobile sensors (tracking) - Cars	0	0	4	33	8	67
Mobile sensors (tracking) - Mobile phone location	0	0	6	38	10	63
Mobile sensors (tracking) - Satellite images	2	15	6	46	5	38
Personal documents	0	0	2	17	10	83
Pictures: Instagram, Flickr, Picasa	3	25	5	42	4	33
Social Networks: Facebook, Twitter, Tumblr etc.	4	25	9	56	3	19
User-generated maps	0	0	6	50	6	50
Videos: Youtube etc.	3	25	5	42	4	33

Comparing Big Data sources mentioned before we observe that data from websites, produced by public agencies and internet searches are easy to obtain in subjective assessment of respondents. Data from social networks and traffic sensors/webcam are sources of medium difficulty to access while data on mobile phone location and credit cards is hard to obtain.

In a way two dimensions of data may be considered: structure and access. It leads us to form the following matrix combining those dimensions for relevant Big Data sources.

Table 3 Dimensions of Big Data sources

		Data structure	
		Structuralized	Non-structuralized
Access	Easy	produced by public agencies	websites, internet searches
	Medium	traffic sensors/webcam	social networks
	Hard	mobile phone location, credit cards	

It reveals our direction of selecting Big Data sources. The top-left corner is the best in both dimensions while bottom-right corner is the worst. As we see the focus should be taken on data produced by public agencies, then on data from traffic sensors/webcam, websites and internet searches. Moreover, sources were assessed with respect to effect on survey conducted by public statistics because we are looking for the source that would decrease the burden of interviewers involved in manual counting of traffic. In results, we found a source of data produced by public agency of medium difficulty of access and adequate database structure – data from traffic sensors gathered by General Directorate of National Roads and Motorways.

Some experience is gained by CBS in dealing with traffic sensors data and we count on support from CBS side.

7.2.6.2. Description of use case

As a result of the inclusion of Poland in the Schengen Information System, on 21 December 2007 the Border Guard discontinued the registration of movement of persons and means of transport at the borders with the countries of the European Union. New regulations on customs clearances resulted in the loss of data on border traffic at the Polish borders with EU member states.

Therefore, methodological work has been undertaken in order to develop methods for collecting the missing data on border traffic. Under the Law on official statistics, since 2014 a sample survey titled “Trips made by non-residents to Poland”. The movement of vehicles and persons at the Polish border with the countries of the European Union has been conducted. The aim of this study is to assess, on a quarterly basis, the number of trips to Poland made by foreign tourists and same-day

visitors, who cross the Polish border with respect to country of origin and destination. Information regarding border traffic is developed based on the results of the aforementioned survey and secure the information needs of official statistics in the field of tourism, balance of payments, national accounts and cross-border areas.

Significant part of that survey is estimation of border traffic intensity with manual counting of vehicles, including plates and type of vehicles. Border traffic is a base for weights calculations. In fact, all of the further results depends on that. Manual counting is very time-consuming while it requires very basic activities. We agreed that all of information gathered by interviewers may be collected without their participation: data on number of vehicles with respect to type may be gathered by road sensor weighting vehicles, country ID from plates can be identify by cameras. We investigated available data in Poland and neighbouring countries from Schengen area.

The data owners in original language are:

- Generalna Dyrekcja Dróg Krajowych i Autostrad for Poland (GDDKiA)
- Bundesanstalt für Straßenwesen for Germany (BASt)
- Ředitelství Silnic a Dálnic for Czech Republic (RSD)
- Národná diaľničná spoločnosť for Slovakia (NDS)
- Kelių ir transporto tyrimo institutas for Lithuania (KTTI)

At this moment we started cooperating with GDDKiA, BASt, KTTI and NDS while we are awaiting the answer from RSD.

GDDKiA is General Directorate of National Roads and Motorways. They conduct two types of measurements called Continuous Traffic Measurement and General Traffic Measurement. Continuous Traffic Measurement is based on over 100 traffic sensors (70 of Golden River Marksman and Verum, 30 of Permanent Enforcement Facility) which collect data 24/7. At this moment the data is available for period 2012-2015. General Traffic Measurement includes manual, semiautomatic and automatic counting of vehicles. Results of that survey are available each 5 years (2005, 2010, 2015) for 1952 points on motorways and 2923 point on national roads. Manual and semimanual counting is a sample survey. The measurement is conducted for 9 or 5 days per year according to traffic intensity observed in previous survey. Annual average daily traffic (AADT) is calculated basing on weighting average of measurements.

BASt is Federal Highway Research Institute and possess over 1500 automatic traffic sensors located on highways and main national roads in Germany. Sensors of our interest are located Saxony, Brandenburg and Mecklenburg-Vorpommern lands. We have already obtained data from Saxony land. Traffic sensors network is so rich that no additional surveys are conducted.

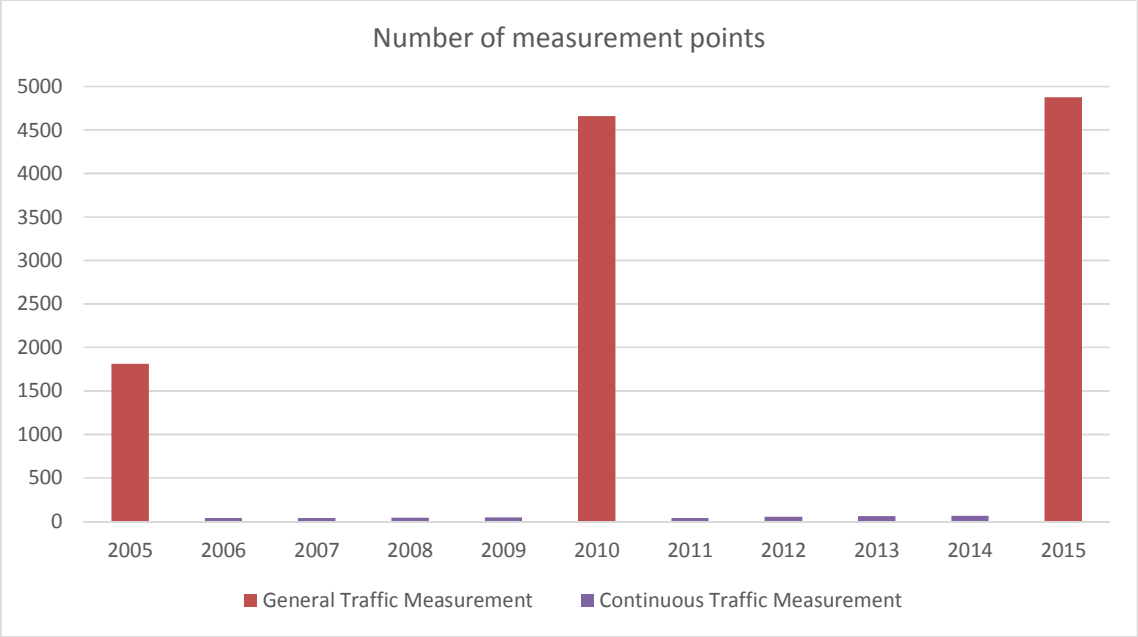
NDS is National Motorway Company in Slovakia. They conduct very similar surveys to GDDKiA. The first one is Nationalwide Traffic Census which took place in 2000, 2005, 2010 and 2015. We have already collected data from that survey for point nearest to Polish-Slovakian border. The second one is data from short-term traffic census with automatic traffic counter for 2013 and 2014 but we have not obtained it yet. They also gather data from electronic toll system.

KTTI is Road and Transport Research Institute in Lithuania. It is a public non-profit seeking legal entity founded for the planning and design of road building, construction, reconstruction, repair and maintenance works and their quality assessment. KTTI's founder and sole shareholder (owner) is a state. KTTI provided monthly data on AADT for 2010-2016 with respect to direction and two types of vehicles.

Simultaneously, there is a sample survey on border traffic conducted by Statistical Office in Rzeszów. We sample days from quarter as well measurements points from all possible measurements points along the border.

The best way to summarize in what situation we are is to interpret the figure presented below.

Figure 1 Number of measurement points in surveys conducted by GDDKiA



Clearly, one the one hand we have large amount of data but only for few points of time while on the other hand we have data of high frequency but of the low volume. Volume of data from GDDKiA is slowly increasing over time. Some of measurement points were not available in the past. Also we have some mirror statistics – from German and Lithuanian side data is of high frequency and plausible volume, for other countries data frequency and volume is available in the same way as for

Poland. In addition, we have data of high frequency for several point near border from our sample survey but we assume that the role of survey will be significantly decreased due to results of our efforts and the data may be used up to certain moment of time in future. After that moment we hope to base only on Big Data sources.

In the first step we prepared a template to fulfill spatial-temporal data. We selected length of time series according to data availability and appropriate measurement points: all points from Continuous Traffic Measurement and all points near the border. Most of the points from Continuous Traffic Measurement is situated in inner area of Poland, only few near the border. For several points mirror statistics are available. Template will be fulfilled for following data sources:

- General Traffic Measurement from GDDKiA
- Continuous Traffic Measurement from GDDKiA
- Continuous Traffic Measurement for German border from BASt
- General Traffic Measurement for Slovakian border from NDS
- Continuous Traffic Measurement for Lithuanian border from KTTI
- Survey on border traffic conducted by Statistical Office in Rzeszów as a support for improving model parameters but not as a source of data for further estimations

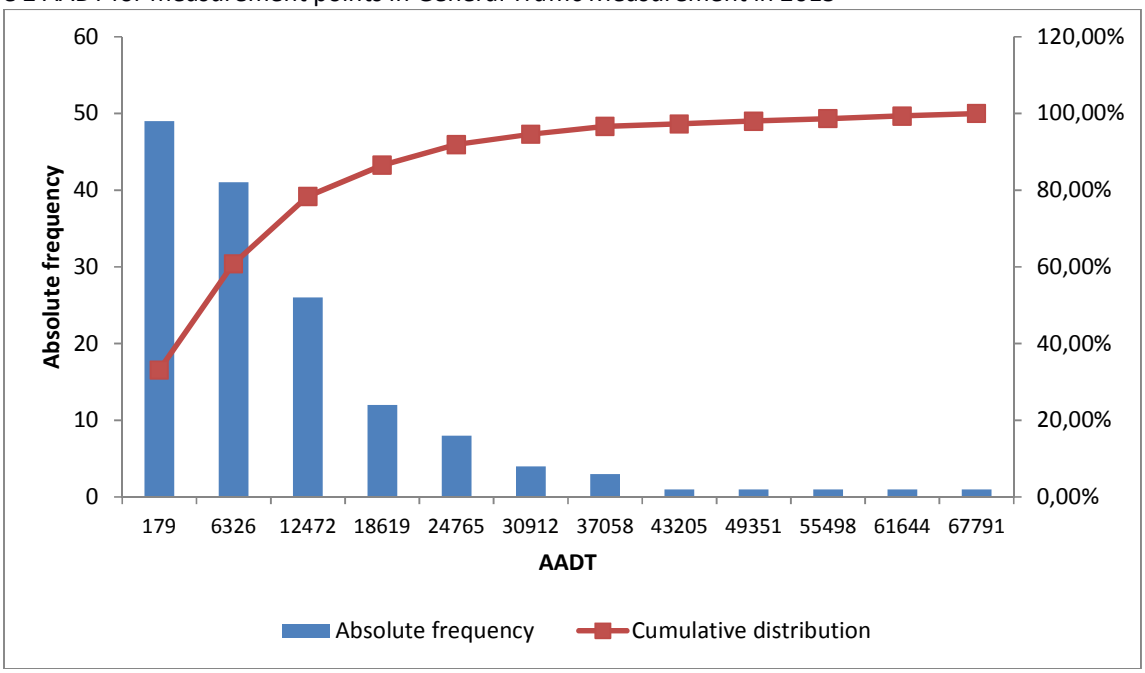
Also additional data will be gathered:

- distance matrix or adjacency matrix between measurement points
- number of registered vehicles in LAU1 level
- other available data which can be connected with traffic

Continuous Traffic Measurement is conducted not for all measurement point for whole period of time – lacks of data at the beginning of time series - as well there are some missing data in the middle of time series. Thus, data will be imputed with use of correlated time series available for the missing moment of time. In a result, we shall obtain the greater set of time series available for the same period.

Although there are some missing data, some spatial and temporal analysis have been already carried out on available data. Figure below presents histogram of AADT.

Figure 2 AADT for measurement points in General Traffic Measurement in 2015



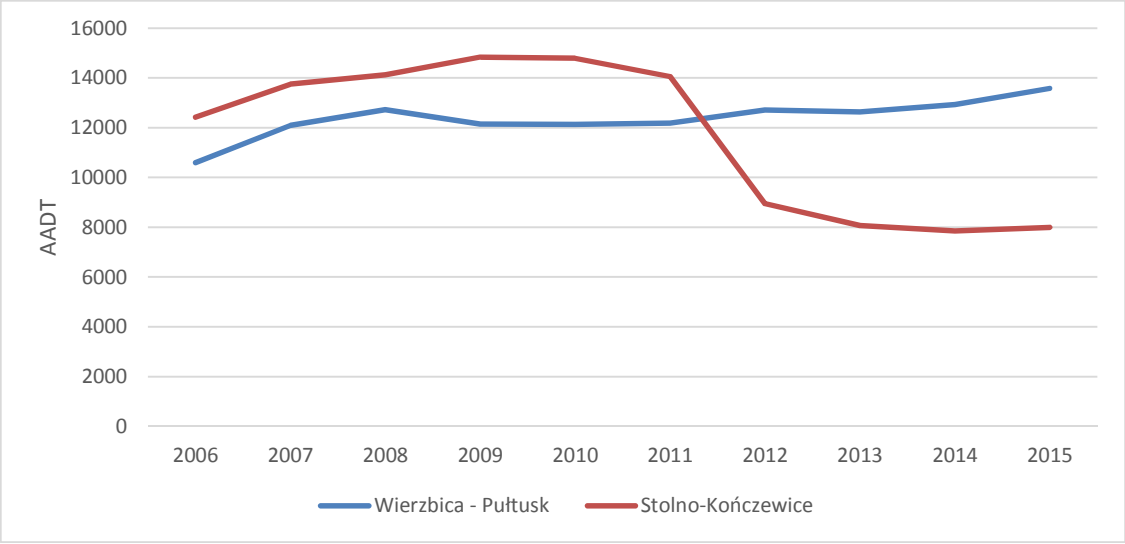
It turns out that traffic intensity is close to follow Pareto rule 80/20. 67% of traffic flows through 32% of measurement points. Distribution is right-skewed and concentrated. We observe a significant traffic intensity dispersion within the most important station points. Basic statistics of spatial distribution are presented below.

Table 4 Distribution of traffic intensity in General Traffic Measurement in 2015

Statistics	Value
Mean	12818
Standard deviation	12329
Coefficient of variation	96%
Median	9622
Kurtosis	5,20
Skewness	1,92
Minimum	179
Maximum	73937
N	148

Basic temporal analysis reveals that most of the time series exhibit nice regularities. Simply, checking R-squared of linear trend model we obtained that more than 47% of time series of relevant length of Continuous Traffic Measurement has R-squared greater than 0,7. That allows us to produce nice forecasts for these time series. On the other hand we discovered that some time series have huge fall (level shift) probably connected with changes in traffic network. Example of well-behaved and ill-behaved time series are presented below.

Figure 3 Traffic intensity at Wierzbica-Pultusk and Stolno-Kończewice points, over the period 2006-2015



There are few crucial steps in building model of traffic intensity which will be performed in our use case:

1) data imputation
2) connecting traffic intensity variables with exogenous variables
3) including distance matrix or adjacency matrix to improve data coherence
4) modelling level shifts
5) building a prior traffic intensity basing on General Traffic Measurements
6) building a posterior traffic intensity basing on Continuous Traffic Measurements from several sources by applying combining data methods
7) temporal disaggregation of yearly data to quarterly data

In fact, steps 1)-3) will be carried out almost simultaneously. First, we shall explore data structure and correlation. Missing data will be imputed using trends, correlation with exogenous variables and traffic time series. Step 4) includes checking connection between observed fall in one time series and rise in other time series. Level falls may modeled with dummy variables and observed level rise in other time series. It might give us a hint how to deal with planned changes in road network before we collect the data on that change.

General Traffic Measurements results from GDDKiA, relevant mirror statistics and exogenous variables will be the base for constructing prior estimation of traffic in measurement points of interest. Although the available data is of high volume. These measurements were taken maximum 3 times between the years 2005 and 2015, every five years. Therefore it will be hard to obtain precise

forecasts and to estimate valuable variance-covariance matrix which includes also points in Continuous Traffic Measurements unless the observed trends are very stable.

In step 6), we assume that data from Continuous Traffic Measurements in Poland and neighboring countries is available in the moment of traffic estimation. Using variance-covariance matrix we shall adjust prior distribution and obtain posterior distribution of traffic intensity. At this stage, methods of combining data shall be involved.

The Last step derives quarterly data from yearly data obtained in previous step. The Temporal disaggregation method will be applied based on data from a survey on border traffic conducted by Statistical Office in Rzeszów and relevant mirror statistics. It may also require combining data methods in a case there are several data sources available.

7.2.7. Combining data

The Non-extensive cross-entropy econometrics model (NCEE) proposed for the Polish – EU border vehicle traffic model (over the period 2005-2016) is intrinsically related to the quality of the initial priors and model data through the Bayesian efficient information optimal processing rules. In the following points, we first describe the data sources for the model and next present the methodology for data combining through the estimation of the Polish – EU border vehicle traffic.

A. Traffic Data production and its characteristics

At the present step of the project, data have been obtained from three sources:

1) Big Data from Polish data owner (GDDKiA),
2) survey results conducted by the Statistical Office in Rzeszów (SOR)
3) Big Data from foreign data owners playing the role of mirror statistics. Data are provided by German, Lithuanian and Slovakian partner. We are looking forward to receiving data soon from the Czech partner.

The survey results from the SOR are obtained following the EU Official Statistics standards. They contain very detailed information in comparison with the Big Data sources.

Polish Big Data from the GDDKiA's sensors remain relatively much more precise. To produce them, the GDDKiA uses automatic, semi-automatic sensors and manual vehicle registration. Automatic sensor is very precise and connected to the system ETC viaTOLL. Without interruption it registers data on vehicle movement along a year period. This system is used for the registration of vehicles on the motorways. The half automatic sensors (Golden River M660/M680, PAT AVC 100, RPP-2, RPP-3, RPP-5) are characterized by lower precision. It combines automatic and manual vehicle

registration. The type of traffic sensor used depends on the specificity of the road segment, while the manual registration can be applied in all kinds of roads.

As Big Data sources, the German, Lithuanian or Slovak traffic sensors are expected to comply with all the EU technical requirements.

Based on the above considerations and those included in the documentations of the above data providers about their different methodologies for the traffic registration, we notice their conformity with the EU traffic registration standards. Thus, Big Data from these sources can be combined in one model without expecting systematic errors from an individual provider.

B. Data combining methodology

1) Data sources to be combined through the NCEE model

We have to estimate the Polish border traffic through internal border of EU over the period 2005-2016, excluding some periods (2005, 2010, 2015) for which the GDDKiA exhaustive database is available. Other data available have been described in the above paragraph: survey data from the SOR, German, Lithuanian and Slovak traffic sensor data.

To better enhance the model estimator precision, new data relating to the properties of the system will be computed. Presently the next two sources of data seem to retain our attention:

a) the scale relation between different border points for the different periods of time of which data are available(2005, 2010, 2015). This scale structure will be estimated under the power law hypothesis, e.g. through the Pareto exponent derivation. This hypothesis is supported by a recent statistical literature about this type of phenomena, but also by the computation outputs carried out by the SOR team.
b) the weights between the Polish internal movement vehicles and the border traffic will be estimated for the periods 2005, 2010 and 2015 and will be interpolated to the remaining periods. We will take into account of the vehicle density structure within the Polish regions being the potential destination of the above border traffic over the period 2005-2015. These data are continuously produced by all the Polish regional departments.

The above available information could be combined through the NCEE model to generate the targeted model outputs (posteriors) based on the prior beliefs and the model data.

Model priors:

since data for the periods 2005, 2010 and 2015 are available, an interpolation (linear or not) will be derived to get the initial structure of Polish-EU border vehicle movement for the period 2005-2015

Model data:

the models restricting the cross-entropy objective function will be proposed based upon formal relations related to the above points a) and b) in B. Data from the survey results conducted by the SOR and the German, Lithuanian and Slovak traffic sensor data will be used in the model through different simulations in accordance with the quality of the model inference information indexes. In spite of the fact these data will be available only for very limited periods, their impact on the whole period outputs may be substantial. This is related to the cross-entropy optimal and systemic properties.

2) The NCEE methodology

i. The **NCEE** rationale:

In line with the rationale for the proposed methodology, the following recent literature-based evidence is non-extensive entropy related:

- The important fact established in recent years is that time-space high frequency (e.g., every minute, hourly, daily) data, that is, Big Data, display Power Law (PL) in most cases. Once aggregated to a lower frequency (e.g., quarterly, yearly) scale, these data in the majority of cases display the Gaussian law, noting a plausible, transitional exponential law.
- Non-extensive entropy, as such, models the non-ergodic systems which compound the Pareto-Levy stable structure converging in the long range to the Gaussian basin of attraction. In the limiting case, non-extensive entropy converges to Shannon- Gibbs entropy, noticing that the maximal magnitude of any random value entropy is achieved under its Gaussian distribution
- PL-related Tsallis entropy should remain, even in the case of a low frequency series, a precious device for econometric modelling since the outputs provided by the exponential family law—e.g., the Gibbs-Shannon entropy approach—correspond to Tsallis entropy, the limiting case when the q-Tsallis parameter equals unity.
- A number of complex phenomena involve long range correlations which can still be seen when data are time-scale aggregated. This is probably because of the interactive nature between the functional relationships describing the involved phenomena and the inheritance properties of power law (PL), or it can depend on their non-linearity. Delimiting the threshold values for a PL transition towards the Gaussian structure (or to the exponential family law) as a function of the data frequency amplitude is difficult since each phenomenon may display its own rate of convergence—if any—towards the central theorem limit attractor.
- Systematic errors from statistical data collecting and processing may generate a kind of tail queue distribution. Thus, a systematic application of the Shannon-Gibbs entropy approach

in the above cases—even on the basis of annual data—could be misleading. In the best case, it can lead to unstable solutions.

- In reverse, since non-extensive Tsallis entropy generalizes exponential family law, the Tsallis-q entropy methodology fits well with high or low frequency series.
- Among the class of a few types displaying higher-order entropy estimators able to generalize the Gaussian law, Tsallis non-extensive entropy has the valuable quality of concavity—and then stability—along the existence interval characterizing most real world phenomena.

ii. The NCEE general formulation:

To better allow the understanding of the procedure, let us present the NCEE model in its general formulation as it follows:

$$\text{(eq.1)}\ Min\ H_q(p//p^0, r//r^0, w//w^0) \equiv$$
$$\alpha \sum p_{km} \frac{\left[p_{km}/p_{km}^0\right]^{q-1} - 1}{q-1} + \beta \sum r_{nj} \frac{\left[r_{nj}/r_{nj}^0\right]^{q-1} - 1}{q-1} + \delta \sum w_{ts} \frac{\left[w_{ts}/w_{ts}^0\right]^{q-1} - 1}{q-1}$$

subject to

$$\text{(eq.2)}\quad Y = X \cdot \beta + e = X \cdot \sum_{m=1}^{M} v_m \left(\frac{p_{km}^q}{\sum_{m=1}^{M} p_{km}^q}\right) + \sum_{j=1}^{J} z_j \left(\frac{r_{nj}^q}{\sum_{j=1}^{J} r_{nj}^q}\right)$$
$$\text{(eq.3)}\quad \sum_{k=1}^{K} \sum_{m>2}^{M} p_{km} = 1$$
$$\text{(eq.4)}\quad \sum_{n=1}^{N} \sum_{j>2}^{J} r_{nj} = 1$$
$$\text{(eq.5)}\quad \sum_{t=1}^{T} \sum_{s>2}^{S} w_{ts} = 1$$

The real q, stands for the Tsallis parameter. Above, $H_q\left(p // p^0, r // r^0, w // w^0\right)$ is nonlinear and measures the entropy in the model. Relative entropies of the three independent systems (the three posteriors p, r and w and the corresponding priors p^0, r^0 and w^0, respectively) are then summed up using the weights α, β, δ. These are real positives summing up to unity under the given restrictions. The symbol $//$ is a "distance metric"[21] of divergence information. We need to find the minimum divergence between the priors and the posteriors while the imposed restrictions must be fulfilled. As will be the case in the application, the first component of the criterion function may concern the parameter structure of the table, the second component errors on column (or row) totals and the last component may concern errors around any additional consistency moment. As has been shown in recent literature, this form of entropy displays the same basic properties as the

21 However, note that K-L divergence is not a true metric since it is not symmetric and does not satisfy the triangle inequality.

traditional Kullback-Leibler relative entropy. The weights α, β, δ could be introduced into the above dual objective function. The first term of "precision" accounts for deviations of the estimated parameters from the prior (generally defined under a support space). The second and the third terms of "prediction ex post" account for the empirical error term as a difference between predicted and observed data values of the model. As expected, the presented entropy model is an efficient information processing rule which transforms, according to Bayes's rule, prior and sample information into posterior information.

As far as the model statistical inference is concerned, from *information properties* and the above formulation of the **NCEE** result that the model inference information indexes should fulfil the basic Fisher-Rao-Cramer information index properties, among them continuity, symmetry, maximum, and additivity.

iii. Combining data through the Polish cross-border traffic NCEE model

The objective function of the NCEE model estimation consists of the minimization of the weighted divergence between unknown posteriors and priors (see eq.1) of a non-extensive Tsallis entropy functional subject to the next Bayesian restrictions:

- Moment formulation in the form of econometric equations from different samples (model data, see eq.2).
- The random component formulation (we will suppose that all available data are known with uncertainty)(see eq.2)
- The regular conditions must sum the probability space points of each parameter up to unity (see eq.3, eq.4 and eq.5).

7.2.8. Summary

Big Data sources that are already obtained are promising in tackling the problem of estimating border traffic. Combining data methods will be crucial in preparing valuable results, especially that there are many mirror statistics available. Methodological ideas shall be useful for many countries in similar situation to Poland. Implementation of pilots result in producing statistics on traffic may decrease the burden of interviewers and increase cost and time efficiency of traffic surveys.

At this stage, we have collected almost all mirror statistics and some additional data for modelling. Next, we will make efforts to complete the set of mirror statistics and enrich the set of auxiliary data relevant to our task. Some statistical analysis and calculations have been already carried out. At this moment, various model assumptions are considered and tested.

Deliverable 7.3. Report on *Agriculture* domain area containing basic information on the data access, quality issues, methodology (focus on combining data) and technical aspects.

7.3.1. Background

Agriculture is one of the sectors of the economy, whose main task is to provide agricultural products. Plant and livestock products are obtained through tillage and plant grown and animal-breeding. Agriculture is also an area which has a strong impact on the environment. In recent decades, the agricultural sector has undergone many changes. At present, all data arising at different stages of agricultural production can be processed and analyzed contributing to increased efficiency, and productivity; or make better use of resources.

Most of the publications or studies (and even the same considerations) about Big Data applications in the context of Agriculture are considered from the perspective of agricultural production. This is because of the benefits for the producer-farmer can bring the appropriate decisions after analyzing available data. In recent years, there have been new phenomena ex.: Internet of Agricultural Things, precision farming/precision agriculture, smart farming (smart analysis and planning, smart control, smart sensing and monitoring, smart logistics), 21st Century Farm.

Manufacturers strive to increase profits and reduce costs, while consumers require healthy and clean food. From the point of view of environmental protection production requires fewer chemicals and less water consumption. Modern agriculture needs, therefore, new products, new methods and technologies, while having regard to issues such as climate change (extreme weather, natural disasters), the limitations of land and water (including as a result of increased urbanization), environmental degradation and changes in nutrition that have a significant impact on health. Big Data comes to meet those needs.

Especially satellite data could be very important for this domain area ex. generating agricultural maps. With successive flights over the same area with a constant interval of time, satellite date allow to monitor changeable field situation. The main satellite data applications in the agriculture domain are as follow: monitoring of crop conditions, seasonal changes, soil properties and mapping tillage activities.

Moreover, satellite data enables us to monitor changes in agricultural production or soil quality and thus to support policy for the sustainable development. Agricultural maps based on satellite data provide independent and objective estimates of cultivation extent in a given country or a growing season.

7.3.2. Data access (with legal and ethical issues)

The free, full and open data policy adopted for the Copernicus programme foresees access available to all users for the Sentinel data[22] products, via a simple pre-registration.

Registration is open to all users via simple on-line self-registration accessible via the Sentinels Scientific Data Hub.

Member States requiring data for national initiatives in the frame of the Sentinels Collaborative Ground Segment need not register on this service, they are served via the dedicated access point.

Following registration, the user can immediately download Sentinel products generated systematically from all acquired data. Please note that depending on the mission and the acquisition time of the product, the full operational qualification may not yet be completed.

The Integrated Administration and Control System (IACS) is a complex administrative information system, mainly used for the purpose of managing payments in the direct support systems, and some of the activities envisaged in the Rural Development Programme (RDP). It was implemented under the provisions of the Act of 18 December 2003 on the National Register of Producers, Register of Farms and Register of Applications for Payment (Journal of Laws of 2012, item 86).

The Land Parcel Identification System (LPIS) forms part of the National Register of Producers, Register of Farms and Register of Applications for Payment (Article 4 in connection with Article 2 of the Act of 18 December 2003 on the National Register of Producers, Register of Farms and Register of Applications for Payment, Journal of Laws of 2004, No. 10, item 76, as amended, uniform text of Journal of Laws of 2012, No. 86). The LPIS is developed on the basis of maps or Land Register documents, as well as other cartographic data. It makes use of various techniques applied within the computer-based geographic information system, including in particular orthoimages or satellite photos, while using a consolidated standard to ensure precision corresponding to at least the cartographic precision of 1:5000.

Implemented by means of the Geo-Information System (GIS) technology, it forms the reference basis for assessing eligibility and inspecting payment applications. It further contains the following regularly-updated digital layers:

- an orthophotomap covering the whole country, with a resolution of 0.5 m and 0.25 m, which is updated in compliance with the applicable EU requirements;

[22] https://sentinel.esa.int/web/sentinel/sentinel-data-access

- land use boundaries, determined using vector data, based on the afore-mentioned orthophotomap;
- referential (cadastral) land lot boundaries, determined using vector data, referred to as reference boundaries which do not have a legal character within the meaning of the geodesy and cartography regulations;
- a processed descriptive land register database, comprising a limited scope of information, as compared to the information falling within starosts' competence;
- vector data serving the purpose of calculating the registration and farming area;
- data from on-site inspections (orthophotomaps, area measurement layers);
- vector data obtained from services and units subordinated to the Minister of the Environment, regarding environmental protection and Natura 2000;
- vector data obtained from the National Water Management Authority, concerning the most vulnerable areas;
- vector data concerning short-rotation coppices, maps of downslope areas, forest areas in the Rural Development Plans for 2004-2006 and 2007-2013, nature monuments, ditches, and a list of areas threatened by water erosion.

LUCAS stands for the Land Use and Coverage Area frame Survey. EUROSTAT has carried out this survey every 3 years since 2006 to identify changes in land use and cover in the European Union. LUCAS surveys are carried out in situ; this means that observations are made and registered in the field all over the EU. The latest published LUCAS survey dates from 2012 and covers all the then 27 EU countries and observations at more than 270 000 points. Eurostat carried out the latest LUCAS survey between March and October 2015. Surveyors from 28 Member States visited a total of 273 401 points[23].

Copernicus is the new name for the Global Monitoring for Environment and Security programme, previously known as GMES. This initiative is headed by the European Commission (EC) in partnership with the European Space Agency (ESA). ESA coordinates the delivery of data from upwards of 30 satellites. The EC, acting on behalf of the European Union, is responsible for the overall initiative, setting requirements and managing the services. The Space Component is managed by ESA and serves users with satellite data available through the Sentinels and the Copernicus Contributing Missions at national, European and international levels. The ground segment, facilitating access to Sentinel and Contributing Mission data, completes the Copernicus Space Component. The Space Component forms the European contribution to the worldwide Global Earth Observation System of Systems (GEOSS). Copernicus provides a unified system through which vast

[23] http://ec.europa.eu/eurostat/web/lucas/overview

amounts of data are fed into a range of thematic information services designed to benefit the environment, the way we live, humanitarian needs and support effective policy-making for a more sustainable future. These services fall into six main categories: land management, the marine environment, atmosphere, emergency response, security and climate change[24].

7.3.3. Methodology

7.3.3.1. Random sampling for the purpose of the in-situ survey

LUCAS points, administrative data coming, e.g., from the IACS (LPIS) system, or other national administrative registers containing spatially-referenced agricultural land use information can be used while selecting the training fields sample. Based on such data, a random or purposive sample can be obtained.

The training fields selection criteria

With a view to identifying the points located within large-scale crop areas, the point location in relation to the parcel layer is additionally analysed. Based on such an analysis, the selected points need to be grouped according to the following criteria:

- the parcel size – depending on the field resolution, the following criteria can be adopted for Sentinel satellite data and administrative data:
    - min. 60x60m fields entirely located within the LPIS parcel,
    - min. 60x60m fields not entirely located within the LPIS parcel,
    - min. 30x30m fields entirely located within the LPIS parcel with an area below 1 ha.

These criteria aim to obtain the highest possible land calibration precision for Sentinel 1 and Sentinel 2 satellite data. However, fields exhibiting smaller spatial values can also be examined using other satellite data or the Copernicus programme which is still under development.

- homogenous crops – enabling the unambiguous crop identification:
    - homogenous crops, which are the subject of detailed description, should be considered any homogenous crops covering an area with the radius of at least 50 m,
    - non-homogenous crops, which do not satisfy the afore-mentioned criterion, should not be included in further analyses.

Based on the reference criteria, a field survey involving interviewers should be conducted, and the implementation period should depend on the plant growth stages. Under Polish climate

[24] http://www.esa.int/Our_Activities/Observing_the_Earth/Copernicus/Overview3

conditions, it should be conducted in May-June. The data collected will serve the purpose of developing crop identification patterns using Sentinel-1 and Sentinel-2 images.

7.3.3.2. Development of remotely sensed data

Multi-temporal satellite data analyses trigger the need to process large sets of images. All the classification-related working stages must be optimised in terms of the operation duration and possible calculations, based on the following principles:

- automated loading of satellite data,
- multi-temporal data processing through segmentation, classification and aggregation.

  Data segmentation can be performed on the basis of:

  - administrative vector data (LPIS),
  - radar satellite images,
  - optical satellite images.

  Depending on the actual data quality and availability, one or several segmentation methods can be used.

  Data classification – it is suggested that the following multi-temporal classifications be used:

  - Support Vector Machine (SVM),
  - Decision Trees (DT),
  - K-Nearest Neighbours (KNN)

  including the following classification parameters:

  - Sigma,
  - Entropy,
  - Alfa,
  - multi-temporal indicators,
  - Wishard distribution.

  Data aggregation – crops can be classified according to the territorial division adopted, making use of the existing administrative division or other aggregates (e.g. a grid).
- results assessment – once processed, the images should be assessed in terms of their usefulness through:
  - analysing the training fields classification error matrix,
  - analysing the calculations accuracy,
  - making comparisons with administrative data,

- making comparisons with statistical data.

7.3.4. Technical aspects

7.3.4.1. Sentinel-1 satellite images

Sentinel satellites have been launched as part of the Copernicus project, which is jointly financed by the European Commission and the European Space Agency (ESA). As a target, a large satellite constellation is to be established, aimed at environmental monitoring, along with the necessary on-the-ground infrastructure. The constellation is expected to comprise around 10 simultaneously-operating satellites. In terms of their functions, the satellites will be divided into five groups, each represented by twin satellites, to ensure high data acquisition frequency. The first group (marked as 1A and 1B) comprises satellites equipped with SARs, while the second group is formed by satellites recording optical images (2A, 2B). Sentinel 1 and 2 are dedicated mainly to land monitoring, whereas ocean monitoring will be conducted by satellites 3A and 3B. Finally, satellites included in Groups 4 and 5 will perform measurements in the Earth's atmosphere, providing real-time data concerning, among others, the chemical composition of the atmosphere.

So far, only the first two satellites, Sentinel-1 and Sentinel-2, have been in operation. Sentinel-1 was launched on 3 April 2014 and was officially put into operation in the autumn 2014, following the trial period. The satellite moves around the polar (helio-synchronous) orbit at the inclination angle of 98° to the equator and at the height of about 700 km. It is equipped with the Synthetic Aperture Radar (SAR) operating at 5,405 GHz. Technical parameters of the radar make it possible to acquire data in four different operation modes:

1. SM (Stripmap Mode) – with 80-km-wide image strips and spatial resolution of 5 x 5 m,
2. IW (Interferometric Wideswath Mode) – with 250-km-wide image strips and spatial resolution of 5 x 20 m,
3. EW (Extra-wide Swath Mode) – with 400-km-wide image strips and spatial resolution of 20 x 40 m,
4. WV (Wave Mode) – collecting data from areas sized 20 x 20 km with spatial resolution of 5 x 5 m, at large inclination angles (23° and 36.5°).

Two polarisation modes, i.e. VH+VV and HV+HH, are available. Sentinel-1A data is currently recorded at six-day intervals. Satellite images obtained through the Sentinel constellation can be accessed free of charge.

7.3.4.2. Sentinel-2 satellite images

Sentinel-2 is a high-resolution multi-spectral satellite offering wide image strips. It was created for the purposes of the Copernicus programme, and in particular to support such measures as plant, water and soil quality monitoring, as well as water-course and coastal zone observation.

Sentinel-2 data is acquired through 13 spectral channels, displaying visible and infrared spectral properties:

- four spectral channels with 10-m spatial resolution: 490 nm (B2), 560 nm (B3),665 nm (B4), 842 nm (B8),
- six spectral channels with 20-m spatial resolution: 705 nm (B5), 740 nm (B6), 783 nm (B7), 865 nm (B8a), 1 610 nm (B11), 2 190 nm (B12),
- three spectral channels with 60-m spatial resolution: 443 nm (B1), 945 nm (B9), 1 375 nm (B10).

Temporal resolution refers to the revisit frequency of a given location. As regards Sentinel-2 data, each satellite offers can monitor the same place at 10-day intervals, which results in temporal resolution of 5 days for the two satellites forming the constellation.

Data acquired through Sentinel-2 satellites, the spatial range of the mission and frequent revisits make it possible to use the devices on a global, regional and local scale. Sentinel-2 data is consistent with data obtained from Landsat and SPOT[25].

7.3.4.3. Software

Multi-temporal coherence matrices for radar satellite images of Poland can be calculated using the MT_SAR programme. It was developed by the Space Research Centre of the Polish Academy of Sciences within the European Space Agency's project No. 4000110261 – *"Feasibility study of land cover classification based on SAR Sentinel-1 images",* implemented between 2014 and 2015. The programmes enables multi-temporal polarimetric transformations using the correlation and coherence matrices that form the basis for polarimetric decomposition which, in turn, is used to model the mechanisms of radar beam reflections on the objects situated on the Earth's surface (land

[25] https://sentinel.esa.int/web/sentinel/user-guides/sentinel-2-msi/overview
https://sentinel.esa.int/web/sentinel/user-guides/sentinel-2-msi/resolutions
https://sentinel.esa.int/web/sentinel/user-guides/sentinel-2-msi/product-types

cover classes). The programme makes use of H/A/Alpha decomposition (Cloude & Pottier, 1997) and is designed to automatically process large image sets.

The programme utilises the following formulas to calculate the coherence matrix for bi-polarised images:

HH/HV polarisation

$$T = \begin{bmatrix} |S_{HH}|^2 & S_{HH}S_{HV}^* \\ S_{HV}S_{HH}^* & |S_{HV}|^2 \end{bmatrix}$$

VV/VH polarisation

$$T = \begin{bmatrix} |S_{VV}|^2 & S_{VV}S_{VH}^* \\ S_{VH}S_{VV}^* & |S_{VH}|^2 \end{bmatrix}$$

HH/HV polarisation

$$k = \frac{1}{\sqrt{2}}[S_{HH} + S_{VV}S_{HH} - S_{VV}]^{\mathrm{T}}$$

$$T = \frac{1}{2}\begin{bmatrix} |S_{HH} + S_{VV}|^2 & (S_{HH} + S_{VV})(S_{HH} - S_{VV})^* \\ (S_{HH} - S_{VV})(S_{HH} + S_{VV})^* & |S_{HH} - S_{VV}|^2 \end{bmatrix}$$

The programme further employs algorithms for calculating the multi-temporal coherence matrix, based on multi-temporal mono- or bi-polarised radar images. The multi-temporal coherence matrix can be calculated for *n* images in the following polarisation patterns: HH, VV, HV or VH:

$$k = \left[S_{HH}^{(1)}S_{HH}^{(2)} \cdots S_{HH}^{(n)}\right]^{\mathrm{T}}$$

$$T = k \otimes k \text{ (Kronecker product)}$$

The function normalising the coherence matrix, in line with the following formula, has also been implemented in the programme:

$$\mathrm{S}_{\mathrm{i}_{\mathrm{norm}}} = \frac{\mathrm{S}_{\mathrm{i}}}{\sqrt{\frac{1}{\mathrm{n}}\sum_{\mathrm{k}=1}^{\mathrm{n}} |\mathrm{S}_{\mathrm{k}}|^2}}$$

where S_{i} values are the constituents of the *n*-dimensional coherence matrix under analysis.

To calculate the multi-temporal H/α decomposition, the following formulas are used:

$$H = -\sum_{k=1}^{n} p_k \log_n p_k$$

where:

λ_k constitute the coherence matrix specific values

$$p_k = \frac{\lambda_k}{\sum_{k=1}^{n} \lambda_k}$$

$$\alpha = \sum_{k=1}^{n} p_k \alpha_k$$

where:

u_{1k} are coherence matrix specific vectors.

$$\alpha_k = \cos^{-1} |u_{1k}|$$

The H/α decomposition calculation can also be based on bi-polarised radar images.

7.3.5. Quality issues

Issues of remote sensing research is interdisciplinary, requiring the knowledge of research teams from different fields of technical and natural sciences. Due to the lack of sufficient knowledge, skills and competences in the resources of public statistics, the solution is cooperation with a specialized research unit, so that the statistics can receive high quality products for further processing and analysis.

At any given time, the satellite data from the Sentinel constellations is characterized by a high level of accuracy, which is available for this type of free of charge data. The size of pixels depends on the type and the mode of the measurement performed by the satellite.

Data delivered upon aggregating the results obtained through the multi-stage processing of satellite images should be validated using reference administrative data and national official statistical data.

7.3.6. Use case

As a first step, an analysis was performed with the aim of identifying the fields potentially situated in agricultural areas. The following vector field development layers (LPIS-based) were used for this purpose:

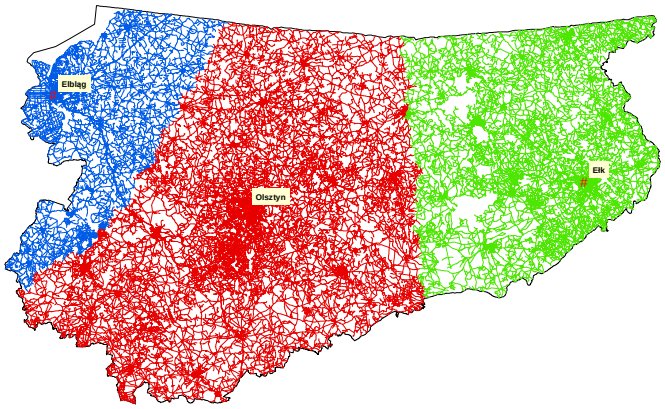
PZ Symbol	Description
Z	Woody or bushy land
I	Other land not suitable for agricultural activity
P	Potentially agricultural land
W	Water
U	Industrial or urbanised area
T	Permanent grassland
S	Orchard
L	Forest
K	Transportation area
D	Habitat

Performance of the in situ survey

Field work was conducted between 15 June and 22 July 2015 and involved four interviewers – two from the Statistical Office in Olsztyn, one from the Statistical Office in Elbląg and one from the Statistical Office in Ełk.

Each interviewer was tasked with visiting all the spatial points assigned to him/her and collecting the necessary data, as specified in the in situ survey form. With a view to optimising the transport time to individual points, network analyses of the transport system in the Warmińsko-Mazurskie Voivodship were conducted, as a result of which network zones were allocated to each of the interviewers involved, and access routes were defined.

The network zones allocation to interviewers in relation to the transport system in the Warmińsko-Mazurskie Voivodship



The interviewers were provided with GPS-equipped tablets enabling them to conveniently reach the allocated points and to take geotagged photos. For each surveyed field, a description had to be prepared, using the pre-defined form which included, among others:

- **Crop type – plant varieties,**
- **Plant growth stage,**

To determine plant growth stages, 10 basic growth and development stages were used (0-9) according to the BBCH (Biologische Bundesantalt, Bundessortenamt and Chemische Industrie) scale:

0 – Germination/Sprouting/Bud development,

1 – Leaf development,

2 – Formation of side shoots,

3 – Stem elongation or rosette growth,

4 – Development of vegetation

5 – Inflorescence emerge,

6 – Flowering,

7 – Development of fruits,

8 – Ripening,

9 – Senescence.

- **Plant measurements,**

  Three to five measurements were performed in the direct vicinity of each surveyed point, in order to determine the average crop height, inter-row width, row spacing and ridge depth.

- **Harvest date,**

  The approximate harvest date was only determined for the period directly preceding the survey.

- **Row image orientation in respect of cardinal points,**

  For each surveyed point, three geotagged photos were taken.

- **Land lay,**

  Land lay description included specifying the plant development conditions resulting, among others, from the exposure to sunlight, wind and soil erosion. Four land lay forms were distinguished:

  - flat,
  - hilly – summit,
  - hilly – slope,
  - hilly – foot.

- **Soil type,**

The soil type was determined by the interviewers on the basis of mechanic soil composition, by squeezing and spreading wet soil lumps in their hands. The soil type classification developed by Schlichting and Blume was used for that purpose, comprising the following categories:

  - sand and loamy sand (a pencil-thick roller cannot be formed),
  - very sandy clay (a half-pencil-thick roller cannot be formed),
  - sandy clay, clay (a half-pencil-thick roller can be formed),
  - clayey loam (when cracked, the soil has a glassy surface),
  - loam (the soil is as soft as butter),
  - other, i.e. turf, alluvial soil, mould.

- **Soil moisture,**

Soil moisture on the survey day was determined using an organoleptic method, referring to a four-degree dampness scale:

  - dry (when touched, sandy and dusty soils are spilled easily, whereas clayey and loamy lumps are hard to crush),
  - fresh (when touched, the soil appears slightly cold and water can be easily absorbed by blotting paper. Sandy and dusty soils do not spread easily, whereas clayey and loamy lumps can be easily crushed but do not smear),
  - damp (the soil appears distinctly cold when touched. When squeezed, water does not leak but is quickly absorbed by blotting paper. The soil sticks to the hand and, when made of a tighter material, e.g. loam, it is easy to roll),
  - wet (the soil smears between the fingers and leaks when squeezed. The wet soil colour is distinctly “cold”, i.e. black, greenish or bluish).

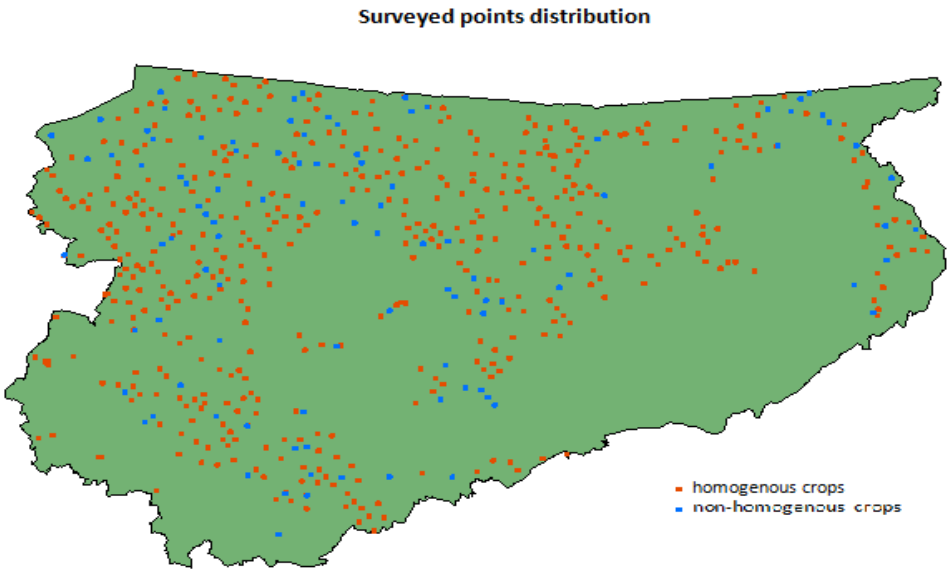
- **Weed infestation,**

Weed infestation is assessed visually using a five-degree scale:

  - no infestation,
  - occasional (weed varieties occurring occasionally),
  - moderate,
  - significant but not dominating,
  - very intensive and crop-suppressing.

The descriptive data gathered during field work was subjected to verification and standardisation. Then, georeferences were established and a geobase was set up. The geotagged images served the purpose of creating an auxiliary file-based geobase. In total, the interviewers described 596 points and took 1847 photos. 495 point situated on cadastral lots occupying a total

area of 143.442019 km2 were described as homogenous crops, whereas 101 points located on cadastral lots with a total area of 14.857946 km^2 were found to correspond to non-homogenous crops.



Reference data

Satellite data was classified using the database containing information about the crops cultivated in the Warmińsko-Mazurskie Voivodship in 2015. The database creation process comprised two stages. First, control areas (lots) were randomly selected, using the LPIS database. Then, a group of interviewers determined the type of crops found in the selected areas. The database delivered to the Space Research Centre of the Polish Academy of Sciences contained geospatial information on the control areas, the cultivated crops and the interviewer's location, along with the questionnaire describing crop and field conditions, and a set of images.

This compilation was developed on the basis of type-2 data, recorded using the VH and VV polarisation pattern. Since the beginning of 2015 over 300 data series have been gathered through Sentinel-1, representing the Warmińsko-Mazurskie Voivodship area. Data has been collected automatically, making use of the functions developed by the Earth Observation Team operating within the Space Research Centre of the Polish Academy of Sciences.

Multi-temporal data processing

Satellite data was classified using the database containing information about the crops cultivated in the Warmińsko-Mazurskie Voivodship.

Coherence matrix calculations

The calculations were performed on the basis of thirteen Sentinel-1 radar images covering the Warmińsko-Mazurskie Voivodship area, recorded between April and October 2015.

The calculations were preceded by processing the satellite data by means of the ESA software provided as part of the Sentinel-1 Toolbox. Intensity and phase calculations, as well as orthorectification, were performed for all images. In addition, as regards the intensity parameter, the Lee filter sized 3 x 3 pixels was applied.

For safety reasons, i.e. to minimise the risk of prolonged calculations, and in order to speed up the calculation process, the images were divided into 42 smaller pieces, taking into consideration the averaging window. In total 546 (13 x 42) images were generated. Following the completion of the calculation process, the fragmented photos were re-arranged into complete images.

Classification of radar beam reflection types using the sorter based on Wishard distribution

The unsupervised polarimetric classification of each of the thirteen images selected was performed using the Sentinel-1 Toolbox (ESA software) functions.

Sigma calculations

Sigma values were calculated for every period covered by the multi-temporal Sentinel-1 data.

Multi-temporal indicator calculations

Multi-temporal intensity data was used in the coherence matrix calculations.

Multi-temporal Sentinel-1 data classification

Segmentation:

Segments were established using optical satellite data gathered through Landsat 8, along with LPIS vector data. To cover the entire surveyed area, it proved necessary to use four different settings characterised by the absence of clouds and acquisition dates meeting the needs of further analyses.

At the first stage, LPIS vector data was used to identify non-agricultural areas (i.e. forests and water bodies). The first segmentation level was created using the LPIS data, corresponding to the LPIS parcel boundaries. The obtained segments were then used at the subsequent segmentation stage, during which the parcel segments were divided in terms of the spectral Landsat image values, so as to ensure that the newly-established segments would accurately reflect the sowing structure.

Classification results

Crops cultivated in the Warmińsko-Mazurskie Voivodship were identified using the eCognition object classification software. The analysis covered only those areas which were defined as used for agricultural purposes in the LPIS database.

Three different classification algorithms were employed, i.e. Decision Trees, K-Nearest Neighbours and Support Vector Machines. The classification included various datasets obtained in the course of analysing the multi-temporal Sentinel-1 dataset, i.e. Sigma, Entropy and Alpha values derived from the coherence matrix, together with multi-temporal indicators and Wishard distribution.

The classification results accuracy was assessed on the basis of an independent dataset containing farmers' declarations gathered by the Agency for Restructuring and Modernisation of Agriculture (ARMA). Table 1 contains the validation results based on an independent ARMA dataset. The most satisfactory results were obtained for the classification established on the basis of the Wishard distribution parameters, using the K-Nearest Neighbours (KNN) method, and on the basis of the time series for the back-reflection coefficient σ0, using the Support Vector Machine (SVM) method.

Table.1. Assessment of the multi-temporal Sentinel-1 radar image classifications based on an independent data set.

No.	Method	Classified data	Total accuracy [%]
1.	Support Vector Machine (SVM)	Sigma	81,0
2.	Support Vector Machine (SVM)	Sigma, Entropy, Alpha	77,5
3.	Decision Trees (DT)	Sigma	73,4
4.	Decision Trees (DT)	Sigma, Entropy, Alpha	73,5
5.	Decision Trees (DT)	Multi-temporal indicators	72,8
6.	K-Nearest Neighbours (KNN)	Wishard distribution	81,7

Following the class combination, an accuracy of 85.6% was obtained for the time series of the back-reflection coefficient σ0 and the Support Vector Machine (SVM) method, while the accuracy for Wishard distribution parameters, using the K-Nearest Neighbours (KNN) method, reached 84.0%.

With a view to assessing the classification results in further detail, the crop areas identified by means of classifying the Sentinel-1 images were compared with the estimated crop areas declared in both the IACS and the annual land use surveys conducted by the Central Statistical Office.

7.3.7. Combining data

What appears essential in terms of using satellite data is the possibility of directly combining it with administrative data. The possibility of directly combining optical and radar data from satellites

contributes to increased accuracy and effectiveness of such application methods. Another advantage of satellite data refers to its use in the current agricultural statistics based on the progressing plant vegetation at a given time.

7.3.8. Summary

The work described in this report involved developing a concept of using radar and optical Sentinel-1 and Sentinel-2 data for the purpose of crop identification. The concept suggests using the time series of crop development data, which should be processed and then classified. Information collected through the in situ survey, and also the existing administrative data, should be used as reference data. The results obtained could be then aggregated in line with the intended spatial division and validated in respect of the reference, administrative and statistical data.

Conclusions

As is clear from the content of this report, during the implementation of SGA-1, there were prepared conditions for carrying out pilot projects. These conditions were aimed at the accuracy of the posed thesis and relate to the possibility of combining different types of Big Data from various sources.

The conditions mentioned above, which are based on the various aspects of Big Data sources (availability, the legality of collecting and processing data, data quality, methodological and IT considerations, etc.) suggest that in the context of separate and described report areas (population, tourism / border crossings and agriculture) will be possible to combine data from different sources and thereby gain valuable data. These data - after confirming their methodological accuracy, will be used to create and enhance new official statistics.

This report summarizes the work what was carried out by two main tasks (TASK- 1 data inventory and TASK 2 - data feasibility) and their results will be an important input to the TASK 3 (Data combination) and TASK 4 (recommendation) that will be carried out under SGA-2.

In the next stage WP 7 will attempt to combine the data beyond current domains defined in project, thus moving to the next level of methodological work in the field of Big Data. For this phase of work, the results of pilot projects will be used, assuming their positive results.

It is planned to cross combine data by schemes:

• Population - Tourism / border crossings,

• Population - Agriculture,

• Agriculture - Tourism / border crossings,

• Population - Tourism / border crossings - agriculture.

Based on the results obtained during the pilot projects, the following opportunities will be checked:

- sentiment survey of the population active in the field of tourism,
- sentiment survey of farmers depending on the crops obtained,
- study of tourism in the group of farmers,
- mix the above themes.

Assuming the pilot projects deliver expected results and there is authorization of data between domains, the trials will be successful. Thus, all the goals of WP 7 will be realized and expanded by additional.

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